Supplementary materials

Contents

1 Introduction

This document presents graphical results generated as output from analytic workflows written in *R* language and available in the public GitHub repository.

2 Species discrimination and prediction

To classify the CHC samples collected from mixed colonies, we need a measure which positions a sample between the *F. sanguinea* and *F. fusca* CHC profiles. Therefore, using `mixOmics` R package [Le Cao et al., 2025], we performed a discriminant analysis training the model on the chemical data collected from the filed colonies in the earlier study [Włodarczyk and Szczepaniak, 2017]. The model was subsequently used to predict identity of new samples.

2.1 Species markers

To figure out the importance of each input variable in the prediction of the species identity made by discriminant model we calculated the weights from the following product matrix [Tenenhaus, 1998, Rohart et al. [2017]].

$$\mathbf{W}(\mathbf{P}^{\top}\mathbf{W})^{-1}\mathbf{C}, \mathbf{P} = \mathbf{XVC} = \mathbf{YV},$$

where
W is the matrix of loading vectors with the number of rows corresponding to the number of input variables and the number of columns corresponding to the number of the latent components
V is the matrix of the coordinates of each observation on latent components
X is the input data matrix

To project the weights assigned to principal components onto the original variables one has to reverse data transformation. Technically, this is achieved by multiplication by pseudoinverse of the rotation matrix produced by PCA. We use pseudoinverse since the rotation matrix has been reduced by removing principal components with low variance.

$$\mathbf{R}^{+}\mathbf{W}(\mathbf{P}^{\top}\mathbf{W})^{-1}\mathbf{C},$$

where $\mathbf{R}^{+}$ denotes the pseudoinverse of rotation matrix from PCA.

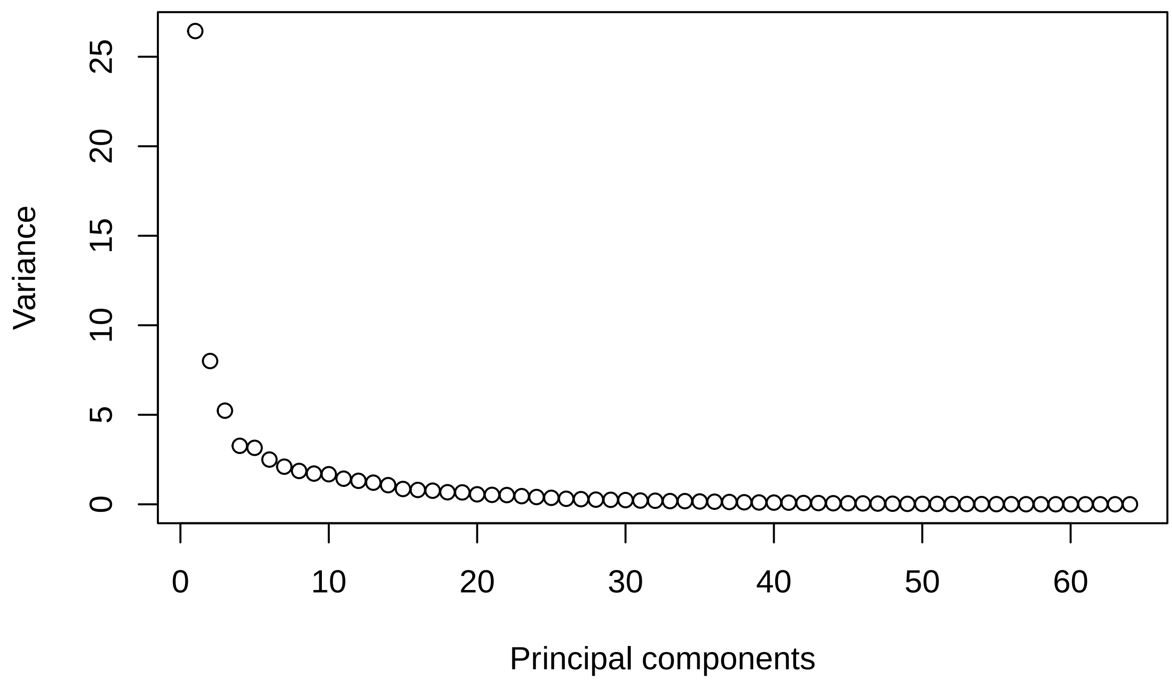


Figure S1 : Variance of each principal compnent. Data was fed into the discriminat model to find features separating both species.

Table S1 : List of the peaks with the species identity score indicating their importance as a marker of *F. fusca* or *F. sanguinea* samples.

Peak ID	Compound	Marker score	fusca marker	sanguinea marker
15	4-MeC24	-0.0250630	TRUE	FALSE
60	13,17-diMeC29	-0.0212807	TRUE	FALSE
14	6-MeC24	-0.0184322	TRUE	FALSE
19	4,12-diMeC24	-0.0183332	TRUE	FALSE
12	3,13-; 3,11-; 3,9-; 3,7-diMeC23	-0.0180813	TRUE	FALSE
10	3-MeC23	-0.0178911	TRUE	FALSE
7	7-Me-C23	-0.0175976	TRUE	FALSE
6	11-Me-C23 + 9-Me-C23	-0.0170052	TRUE	FALSE
33	6-MeC26	-0.0168284	TRUE	FALSE
37	7,x-; 5,x-; 10,14-diMeC26	-0.0165982	TRUE	FALSE
31	3,13-diMeC25	-0.0162913	TRUE	FALSE
32	14-; 10-MeC26 + 3,7,11-triMeC25	-0.0157412	TRUE	FALSE

Table S1 : List of the peaks with the species identity score indicating their importance as a marker of *F. fusca* or *F. sanguinea* samples. *(continued)*

Peak ID	Compound	Marker score	fusca marker	sanguinea marker
13	12-; 11-; 10-; 9-; 8-MeC24	-0.0154182	TRUE	FALSE
49	7,11-diMeC27	-0.0141977	TRUE	FALSE
40	4,12-diMeC26	-0.0137364	TRUE	FALSE
44	7-MeC27	-0.0136976	FALSE	FALSE
22	13-; 11-; 9-MeC25	-0.0131324	FALSE	FALSE
21	4,12,16-triMeC24	-0.0129441	FALSE	FALSE
69	C31	-0.0128177	FALSE	FALSE
11	C24	-0.0123867	FALSE	FALSE
27	7,11-diMeC25 + 3-MeC25	-0.0122914	FALSE	FALSE
5	C23	-0.0117915	FALSE	FALSE
8	5-Me C23	-0.0110041	FALSE	FALSE
57	4,8,12-triMeC28	-0.0107913	FALSE	FALSE
3	C23-ene	-0.0101536	FALSE	FALSE
42	4,8,14-MeC26	-0.0100388	FALSE	FALSE
55	C29ene	-0.0083950	FALSE	FALSE
26	11,15-; 9,13-diMeC25	-0.0080903	FALSE	FALSE
34	5-MeC26	-0.0074724	FALSE	FALSE
65	3,7-; 3,9-; 3,11-diMeC29	-0.0073725	FALSE	FALSE
16	10,14-diMeC24	-0.0064382	FALSE	FALSE
63	C30ene	-0.0044467	FALSE	FALSE
53	14-; 12-; 10-MeC28	-0.0035012	FALSE	FALSE
23	9-MeC25	-0.0031749	FALSE	FALSE
24	7-MeC25	-0.0026105	FALSE	FALSE
77	C33ene	-0.0004307	FALSE	FALSE
47	9,17-; 9,15-diMeC27	-0.0003866	FALSE	FALSE
56	C29	0.0015481	FALSE	FALSE
20	C25	0.0018378	FALSE	FALSE
51	C28 + 3,15-diMeC27	0.0020822	FALSE	FALSE
58	15-; 13-; 11-; 9-; 7-MeC29	0.0021725	FALSE	FALSE
43	13-MeC27	0.0029545	FALSE	FALSE
28	5,13-diMeC25	0.0033343	FALSE	FALSE
59	11,17-diMeC29	0.0041631	FALSE	FALSE
9	9,13-diMeC23	0.0044141	FALSE	FALSE
48	9,13-diMeC27	0.0051678	FALSE	FALSE
66	14-; 13-; 12-; 11-MeC30	0.0052522	FALSE	FALSE
4	C23-ene	0.0058229	FALSE	FALSE
18	C25-ene	0.0070844	FALSE	FALSE
52	3,9-; 3,7-diMeC27	0.0074879	FALSE	FALSE
36	10,16-; 10,14-diMeC26	0.0091723	FALSE	FALSE
72	13,19-;13,17-diMeC31	0.0096029	FALSE	FALSE
80	11,21-; 11,19-diMeC33	0.0101107	FALSE	FALSE
35	4-MeC26	0.0108916	FALSE	FALSE
54	8,12,16-triMeC28	0.0110194	FALSE	FALSE
62	5,9-; 5,11-; 5,13-diMeC29	0.0113611	FALSE	FALSE
75	x-Me_C32	0.0114659	FALSE	TRUE
30	C26	0.0121636	FALSE	TRUE

Table S1 : List of the peaks with the species identity score indicating their importance as a marker of *F. fusca* or *F. sanguinea* samples. *(continued)*

Peak ID	Compound	Marker score	fusca marker	sanguinea marker
79	x-Me_C32	0.0131711	FALSE	TRUE
17	C25-ene	0.0142664	FALSE	TRUE
74	(di)MeC31, mix	0.0152084	FALSE	TRUE
70	15-; 13-; 11-; 9-MeC31	0.0152506	FALSE	TRUE
76	13,17-DiMe_C32 + x,y-DiMe_C32	0.0153961	FALSE	TRUE
46	11,15-diMeC27	0.0158363	FALSE	TRUE
45	5-MeC27	0.0160579	FALSE	TRUE
38	C27-ene + 6,12-diMeC26	0.0180686	FALSE	TRUE
71	11,19-; 11,17-; 11,15-diMeC31	0.0186094	FALSE	TRUE
41	C27	0.0208482	FALSE	TRUE
67	(di)MeC30 (mix)	0.0283672	FALSE	TRUE
25	5-MeC25	0.0294244	FALSE	TRUE
50	C28ene	0.0315389	FALSE	TRUE

2.2 Species identity score

Ants in the mixed colonies exchanged their CHCs to produce a hybrid chemical identity. We developed a method to assign each sample a value measuring similarity to one or the other species. It is based on a discriminant model that predicts the class of new samples using `predict` method from `mixOmics` R package. The input data are processed in the same way as the training set, i.e. central-log transformation is followed by projection into principal component space. In the latter step, the same rotation matrix and scaling vector is applied as those used for the training set are applied. This is achieved by using `predict` method on the object of `prcomp` class.

2.2.1 Predicted species of different categories of ants as a function of the proportion of *F. sanguinea* ants in a colony

The Species Identity Score was computed for *F. sanguinea* and *F. fusca* samples from mixed colonies and used as a response variable in linear mixed models. As a fixed effect we used the proportion of *F. sanguinea* workers among all workers in a colony. Colony ID and sampling occasion were incorporated as random effects.

In the following model specifications (1|random_factor) denotes random intercept (separate for each level of the random factor) and ((1|random_factor_1:random_factor_2)) denotes random intercepts generated from the combinations of two factors. β_0 and ϵ denote intercept and error term, respectively. Included are summary reports, results of the Shapiro-Wilk test of nornalizty of conditional residuals, as well as diagnostic plots and tests generated with the use of `DHARMa` R package[Hartig, 2024]. The response variable often needed to be transformed to meet model assumptions. Random terms with no variance have been dropped.

2.2.1.1 Mature *F. sanguinea*

$$\sqrt{\text{Species\_Identity\_Score}} = \beta_0 + \text{sanguinea\_proportion} + (1|\text{colony\_ID:sampling\_occasion\_ID}) + \epsilon,$$

where Species_Identity_Score denotes the Species Identity Score of the mature *F. sanguinea* workers.

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
```

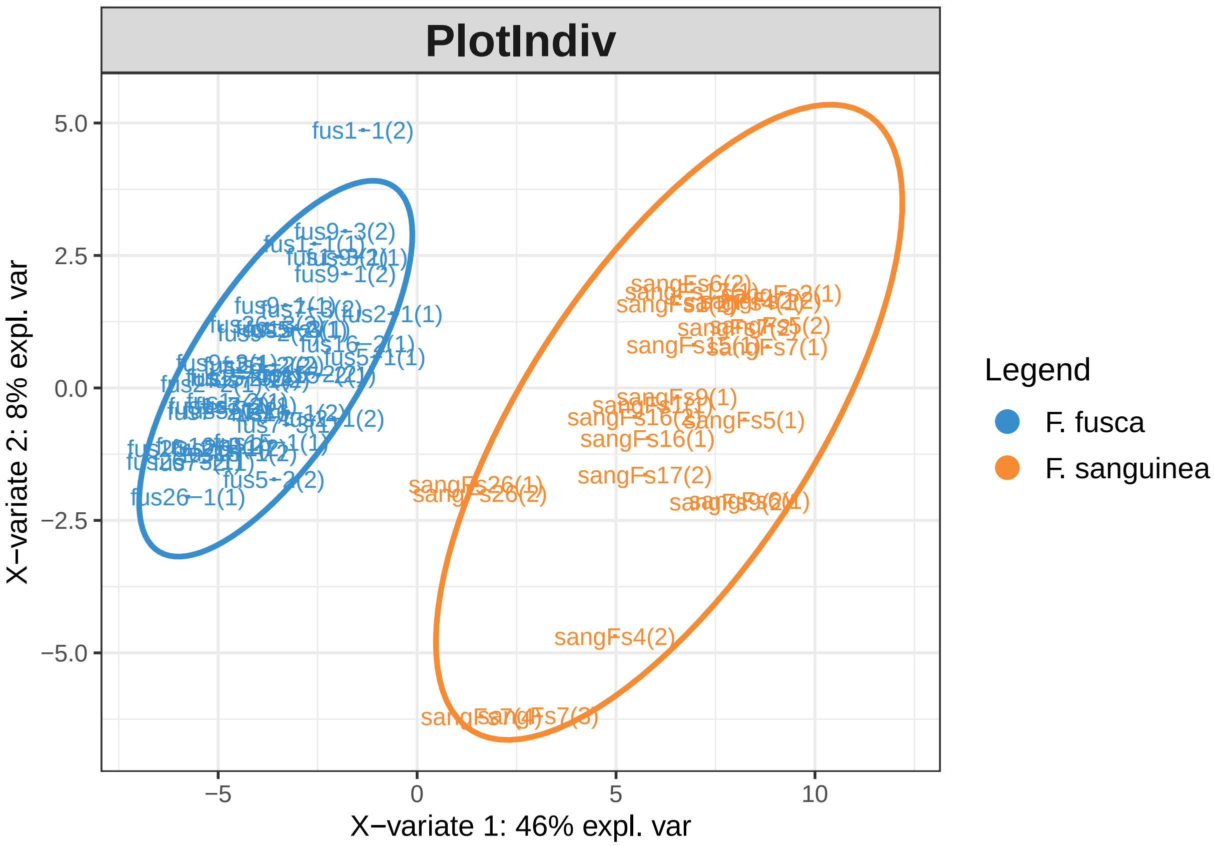


Figure S2 : Distribution of the samples projected onto two first latent components before optimizing the number of model paramaters.

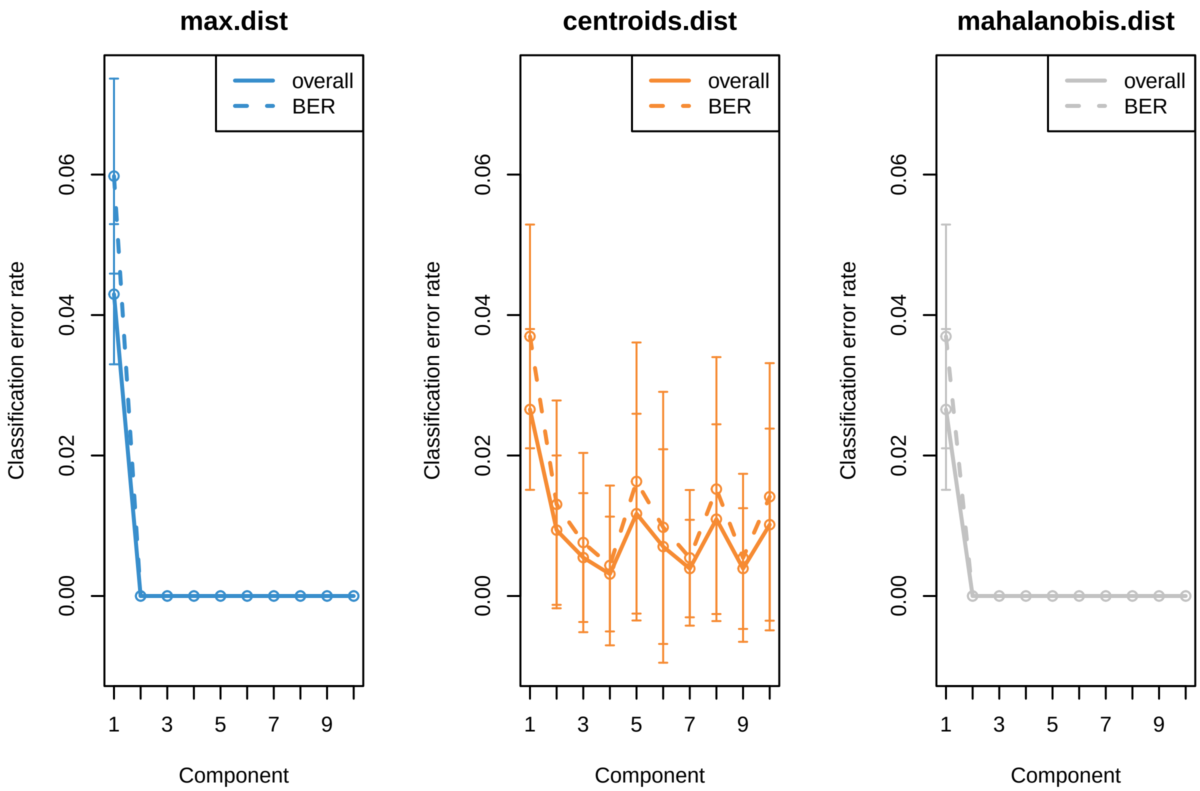


Figure S3 : Results of the performance test of the discriminant model with different number of the latent components. For details see the documunetation of R mixOmics package.

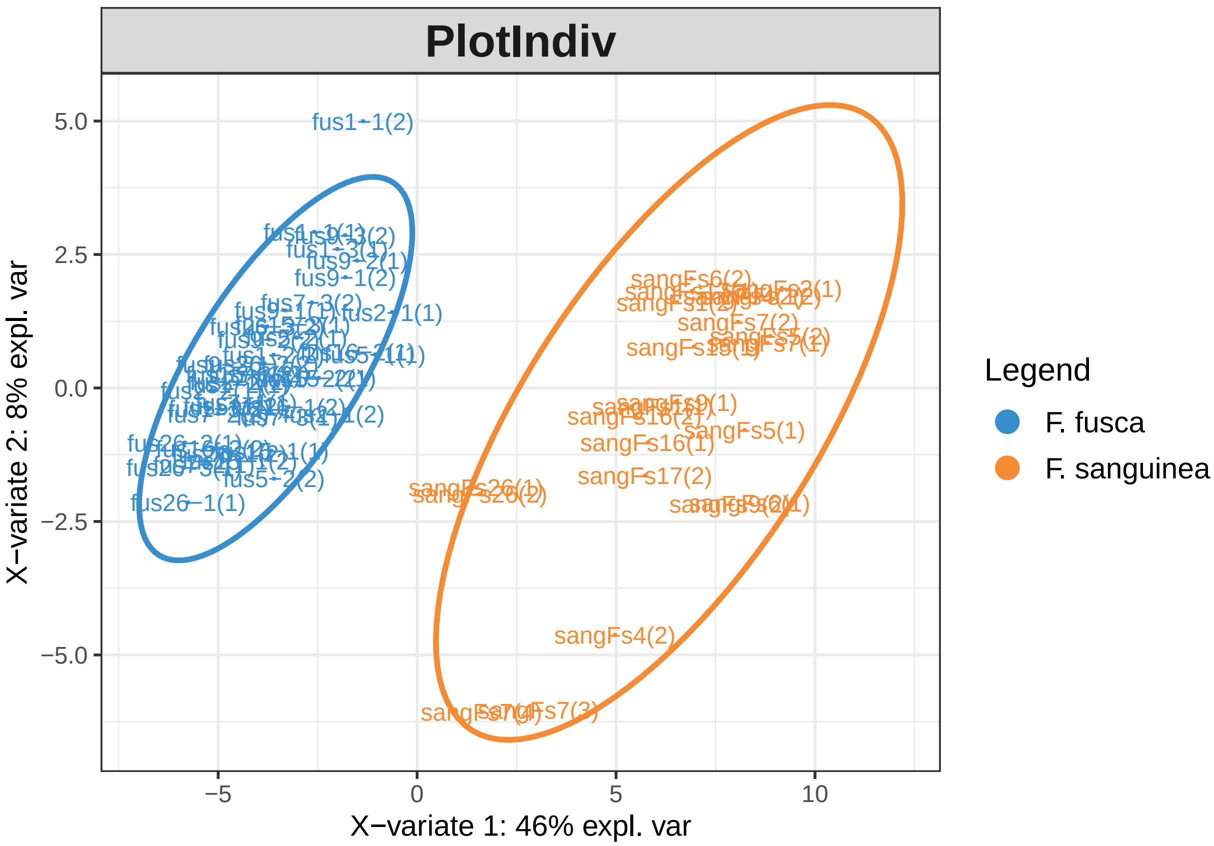
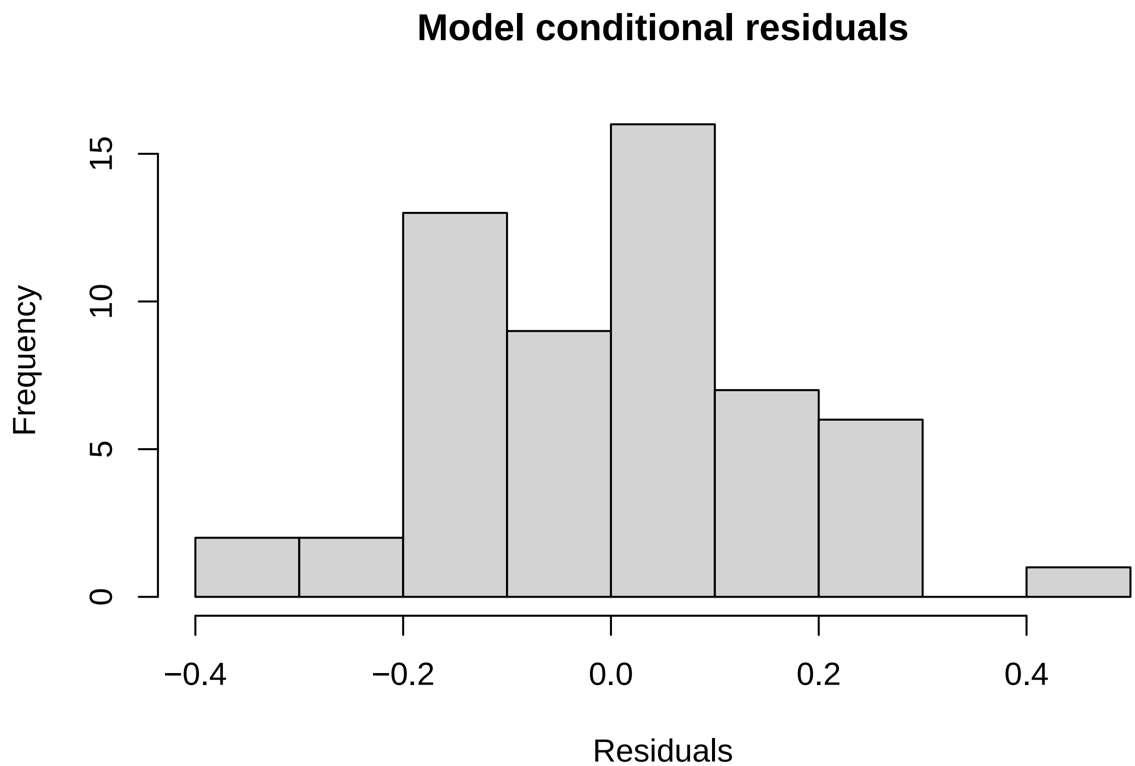
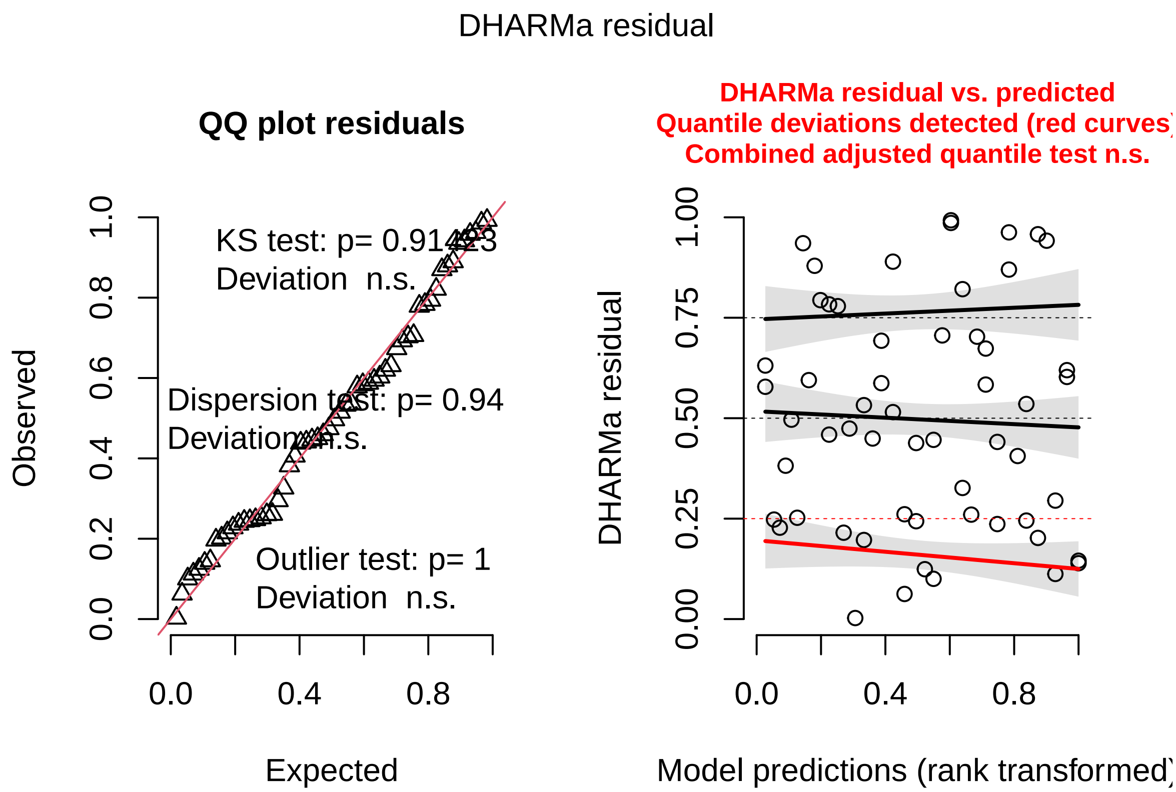


Figure S4 : Projection of the samples in onto two first discriminant analysis components after model tuning.

```
## Formula: I(sqrt(predicted_species)) ~ sang_prop + (1 | colony:census_date)
##    Data: model_input
##
## REML criterion at convergence: 21.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.94450 -0.55347  0.05334  0.44781  2.07622
##
## Random effects:
##  Groups             Name        Variance Std.Dev.
##  colony:census_date (Intercept) 0.04510  0.2124
##  Residual                       0.04203  0.2050
## Number of obs: 56, groups:  colony:census_date, 39
##
## Fixed effects:
##             Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)  0.68388    0.08465 39.36235   8.079 6.95e-10 ***
## sang_prop    1.04551    0.15357 36.97695   6.808 5.13e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr)
## sang_prop -0.852
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.98654, p-value = 0.7851
```


Model conditional residuals
Frequency
0
5
10
15
−0.4
−0.2
0.0
0.2
0.4
Residuals



2.2.1.2 *F. fusca*

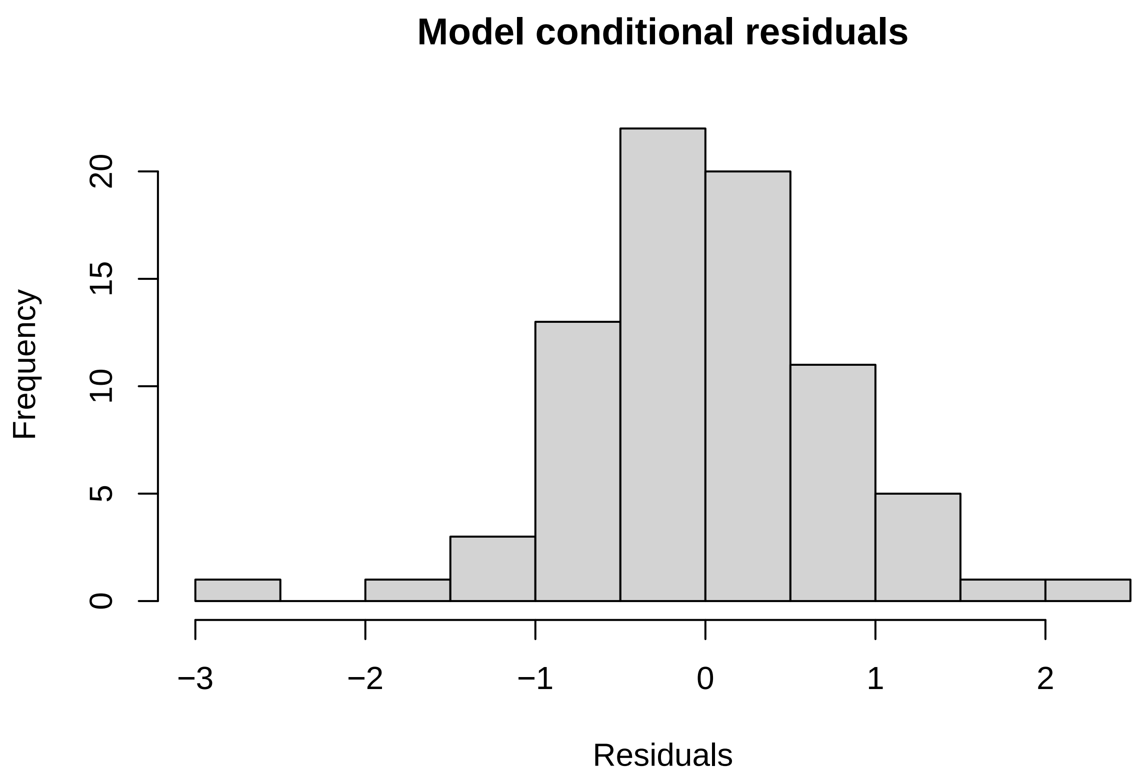
$$\text{Species\_Identity\_Score} = \beta_0 + \text{sanguinea\_proportion} + (1|\text{colony}) + (1|\text{colony\_ID:sampling\_occasion\_ID}) + \epsilon,$$

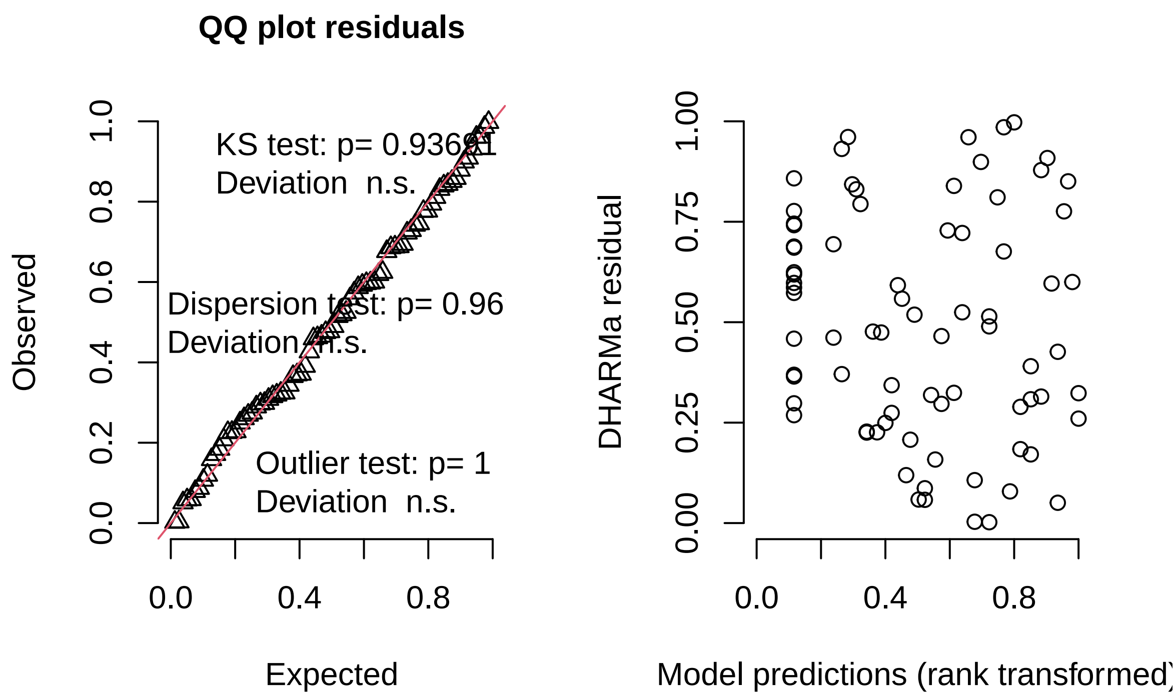
where Species_Identity_Score denotes the Species Identity Score of the mature *F. fusca* workers.

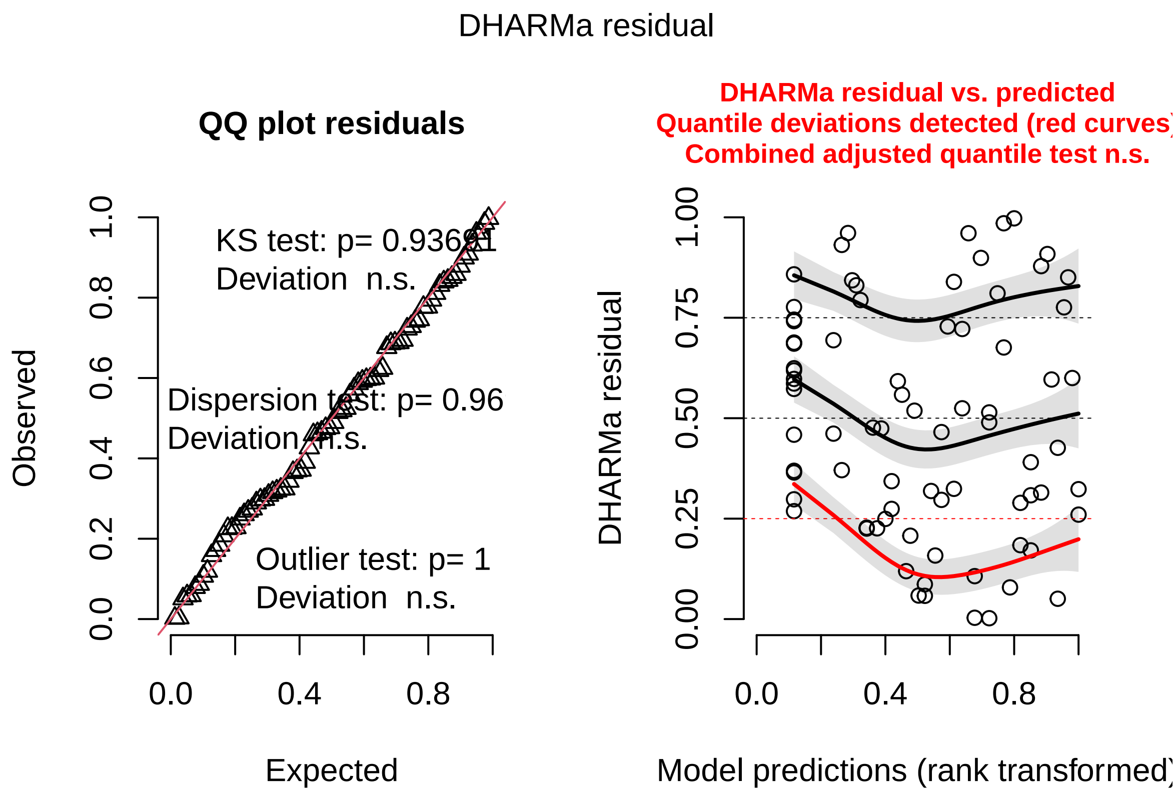
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: predicted_species ~ sang_prop + (1 | colony) + (1 | colony:census_date)
##    Data: model_input
##
## REML criterion at convergence: 232.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.85500 -0.48958 -0.00767  0.51370  2.27141
##
## Random effects:
##  Groups             Name        Variance Std.Dev.
##  colony:census_date (Intercept) 0.2851   0.5340
##  colony             (Intercept) 0.1559   0.3949
##  Residual                       0.7934   0.8908
## Number of obs: 78, groups:  colony:census_date, 55; colony, 20
##
## Fixed effects:
##             Estimate Std. Error      df t value Pr(>|t|)
```

```
## (Intercept)  -1.1692     0.2072 42.0820  -5.644 1.28e-06 ***
## sang_prop     3.5082     0.4493 38.0222   7.807 2.04e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr)
## sang_prop -0.653
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.98166, p-value = 0.3235
```




QQ plot residuals
KS test: p= 0.93691
Deviation n.s.
Dispersion test: p= 0.96
Deviation n.s.
Outlier test: p= 1
Deviation n.s.
Observed
Expected
DHARMa residual
Model predictions (rank transformed)
0.0
0.2
0.4
0.6
0.8
1.0
0.00
0.25
0.50
0.75
1.00



2.2.1.3 Callow *F. sanguinea*

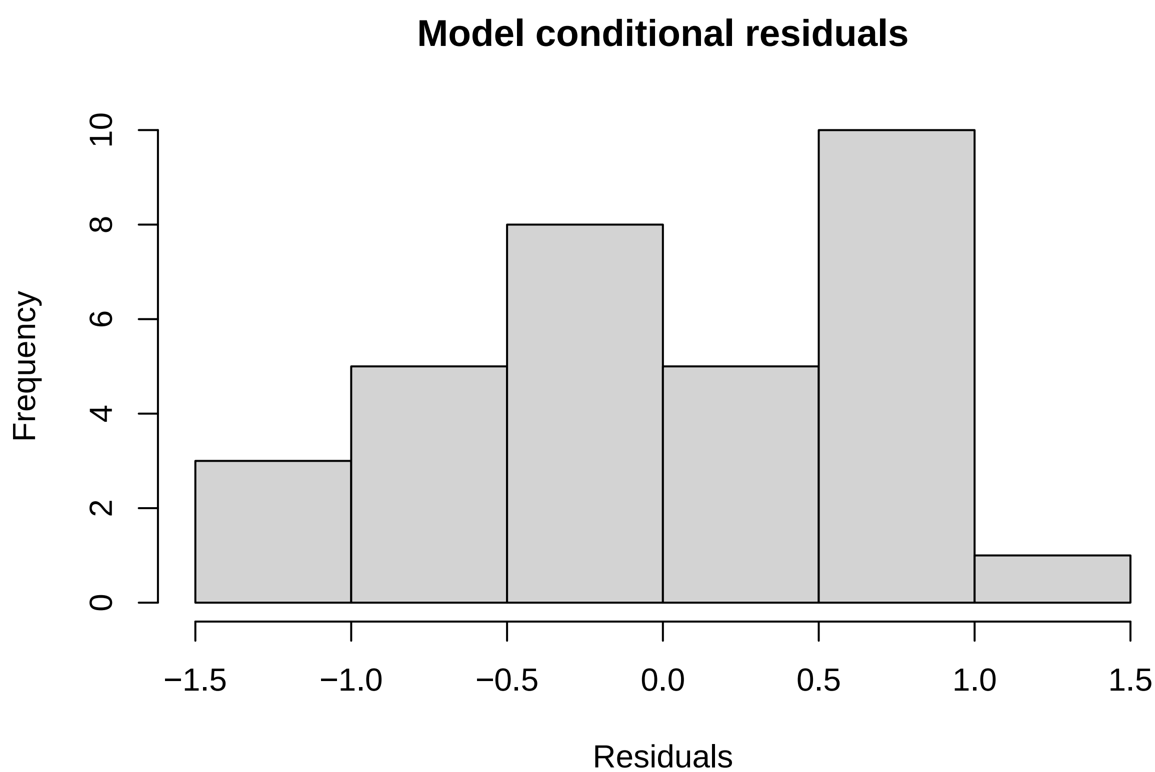
$$\text{Species\_Identity\_Score} = \beta_0 + \text{sanguinea\_proportion} + (1|\text{colony}) + \epsilon,$$

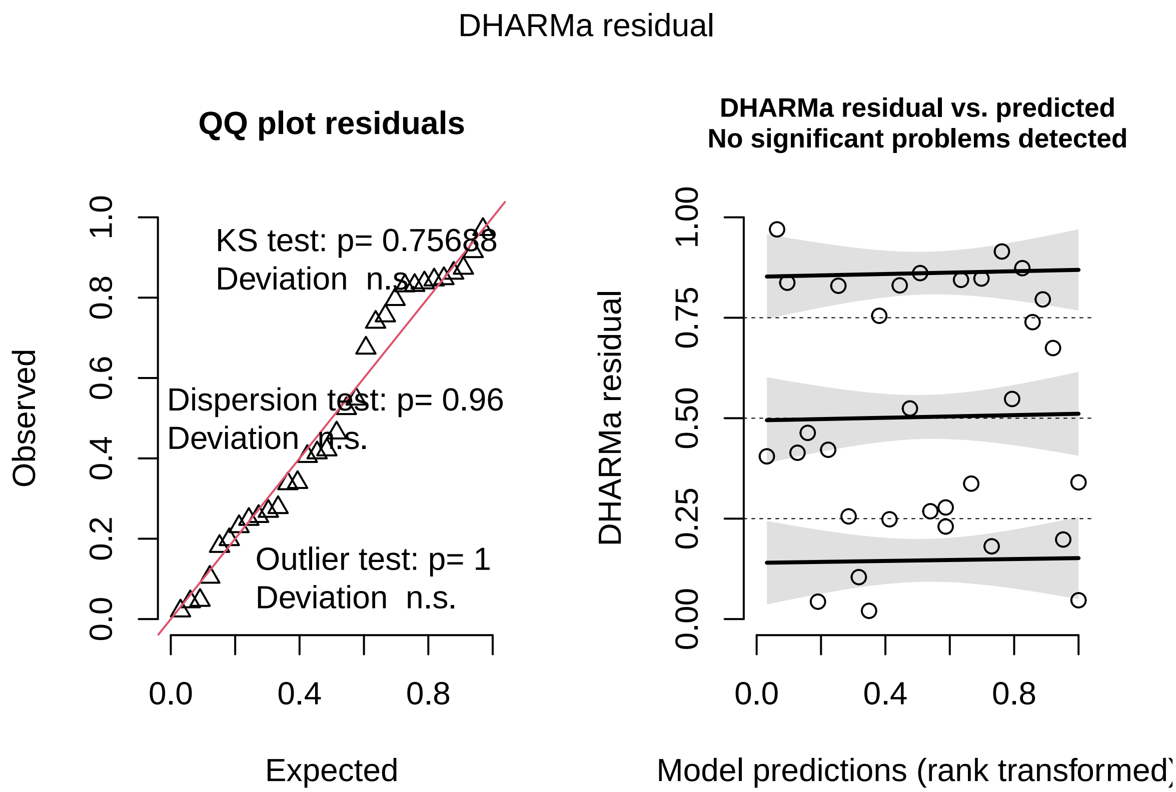
where Species_Identity_Score denotes the Species Identity Score of the callow *F. sanguinea* workers.

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: predicted_species ~ sang_prop + (1 | colony)
##    Data: model_input
##
## REML criterion at convergence: 75.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.99822 -0.59458 -0.05828  0.79454  1.63728
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  colony   (Intercept) 0.08082  0.2843
##  Residual             0.55288  0.7436
## Number of obs: 32, groups:  colony, 16
##
## Fixed effects:
##             Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  -0.5972     0.2529 28.9183  -2.361   0.0252 *
```

```
## sang_prop     2.5205     0.4622 28.4078   5.454 7.68e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr)
## sang_prop -0.798
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.9667, p-value = 0.4136
```

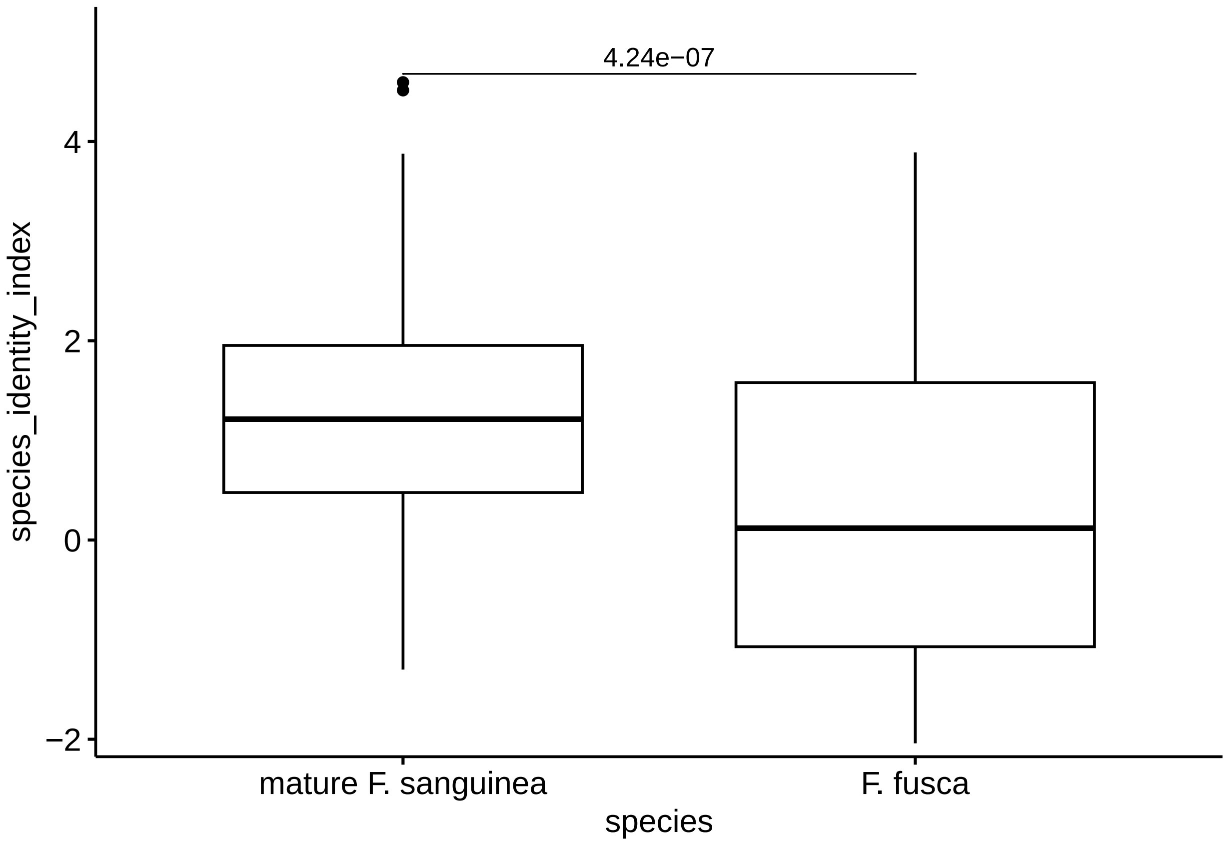




2.2.2 Difference in Species Identity Score between mature *F. sanguinea* ants and *F. fusca* slaves

Since the linear model does not meet one of diagnostics criteria, a Wilcoxon paired test was applied.

```
##
##  Wilcoxon signed rank test with continuity correction
##
## data:  model_input$predicted_species and model_input$slave_species_index
## V = 1847, p-value = 4.236e-07
## alternative hypothesis: true location shift is not equal to 0
```



2.2.3 Difference in Species Identity Score between callow *F. sanguinea* ants and *F. fusca* slaves

$$\text{SIS\_diff} = \beta_0 + \text{sanguinea\_proportion} + (1|\text{colony}) + \epsilon,$$

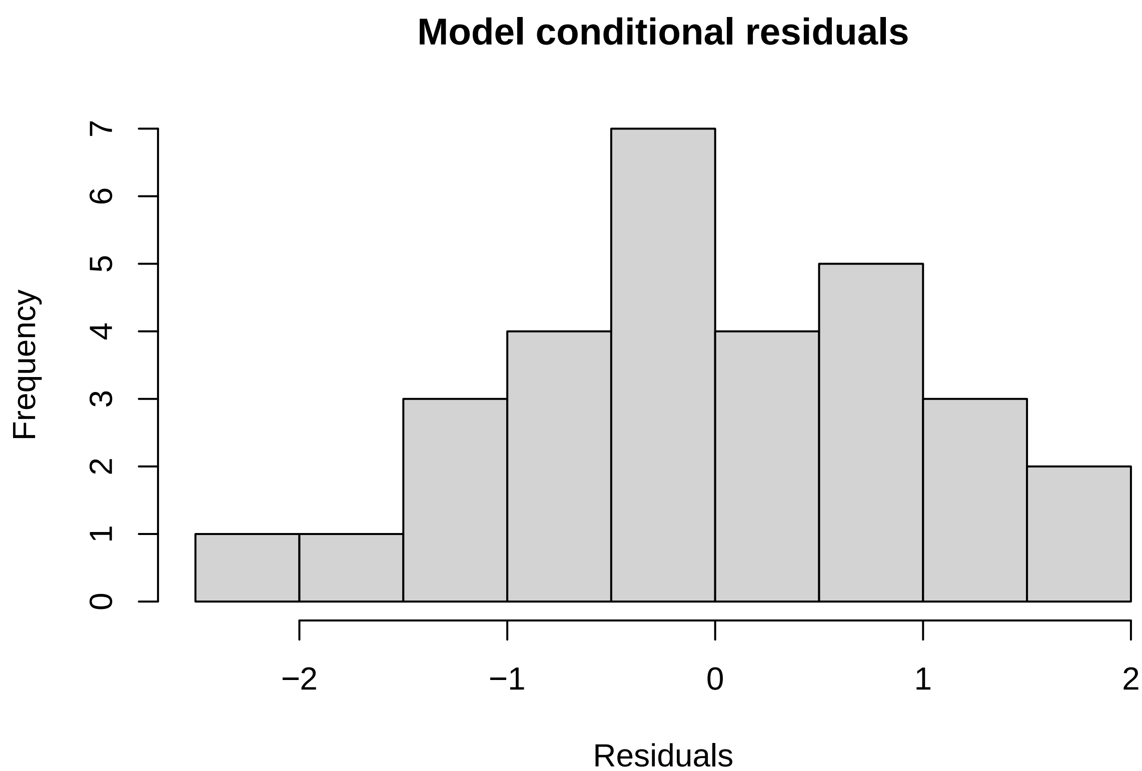
where SIS_diff denotes the difference in Species Identity Score between the mature *F. sanguinea* and *F. fusca* sampled at the same time and from the same colony. Replicated samples were averaged before the calculations.

Since the linear model does not meet one of diagnostics criterion we will apply Wilcoxon paired test.

```
##
## Call:
## lm(formula = index_diff ~ sang_prop, data = model_input)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.4202 -0.6187 -0.1089  0.7401  1.7652
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.4089     0.3375   1.211    0.236
## sang_prop    -0.6396     0.6960  -0.919    0.366
##
## Residual standard error: 1.029 on 28 degrees of freedom
##   (2 observations deleted due to missingness)
```

```
## Multiple R-squared:  0.02928,    Adjusted R-squared:  -0.005391
## F-statistic: 0.8445 on 1 and 28 DF,  p-value: 0.366
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.97689, p-value = 0.7383
```

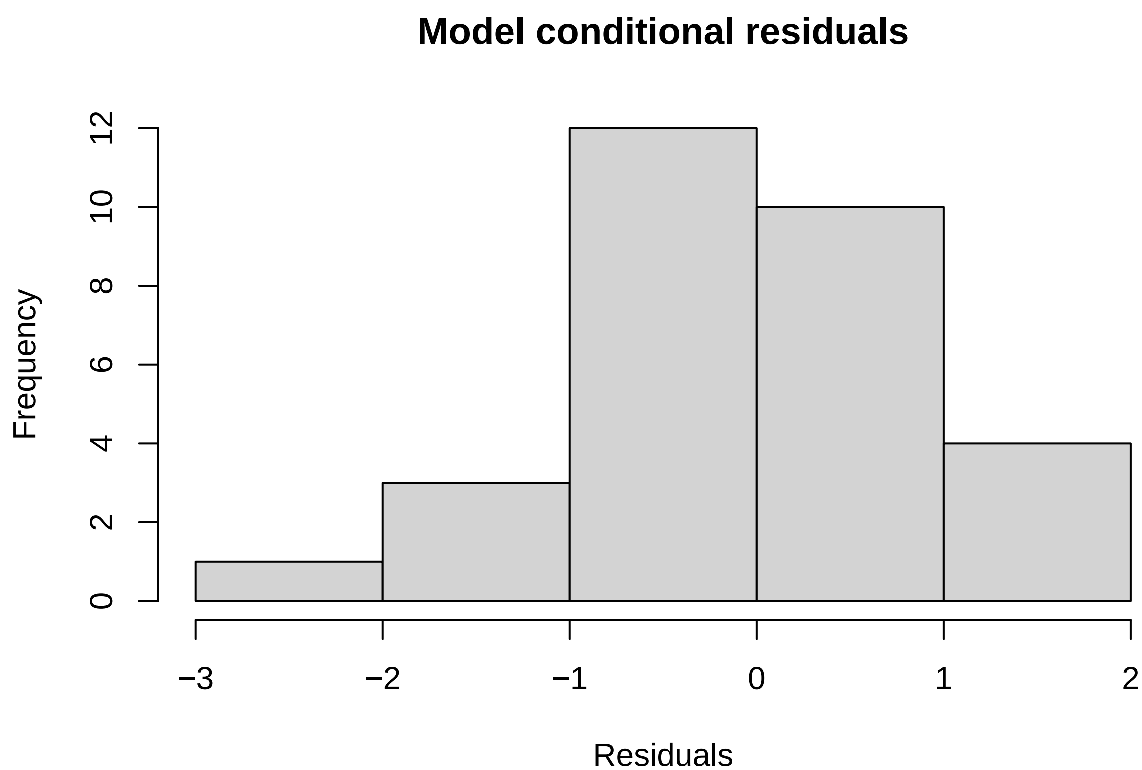


2.2.4 Difference in Species Identity Score between callow and mature *F. sanguinea* ants.

```
##
## Call:
## lm(formula = index_diff ~ sang_prop, data = model_input)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.5102 -0.5283 -0.1241  0.6490  1.8962
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.7018     0.3245  -2.162   0.0393 *
## sang_prop    -0.1626     0.5937  -0.274   0.7861
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 0.9691 on 28 degrees of freedom
##   (2 observations deleted due to missingness)
## Multiple R-squared:  0.002673,   Adjusted R-squared:  -0.03295
## F-statistic: 0.07504 on 1 and 28 DF,  p-value: 0.7861
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.98353, p-value = 0.9096
```



3 Finding peaks differentiating callow and mature *F. sanguinea* workers

The procedure identifying species markers was also applied to determine peaks characteristic of *F. sanguinea* callow ants. In discriminant analysis, samples were classified into one of the three groups: callow *F. sanguinea*, adult *F. sanguinea*, or adult *F. fusca*.

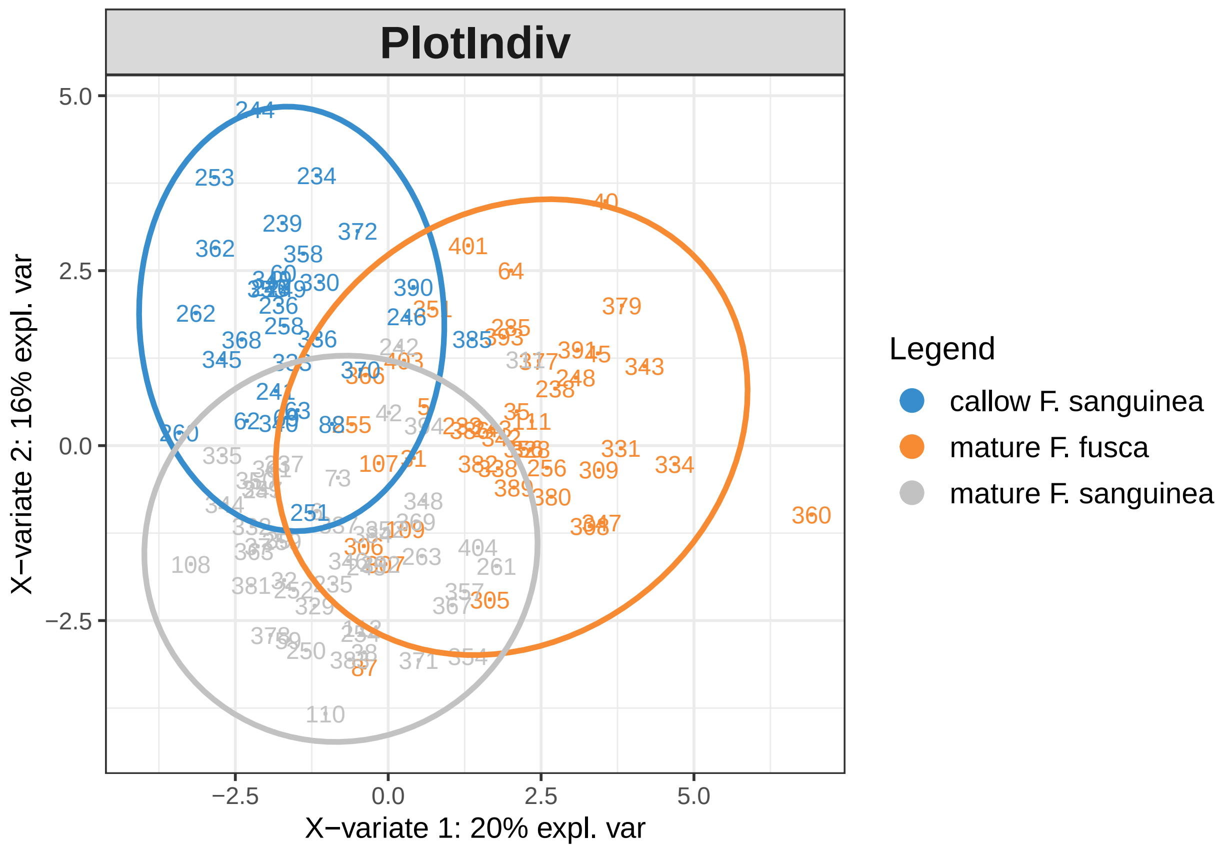


Figure S5 : Distribution of the samples projected onto two first latent components before optimizing the number of model paramaters.

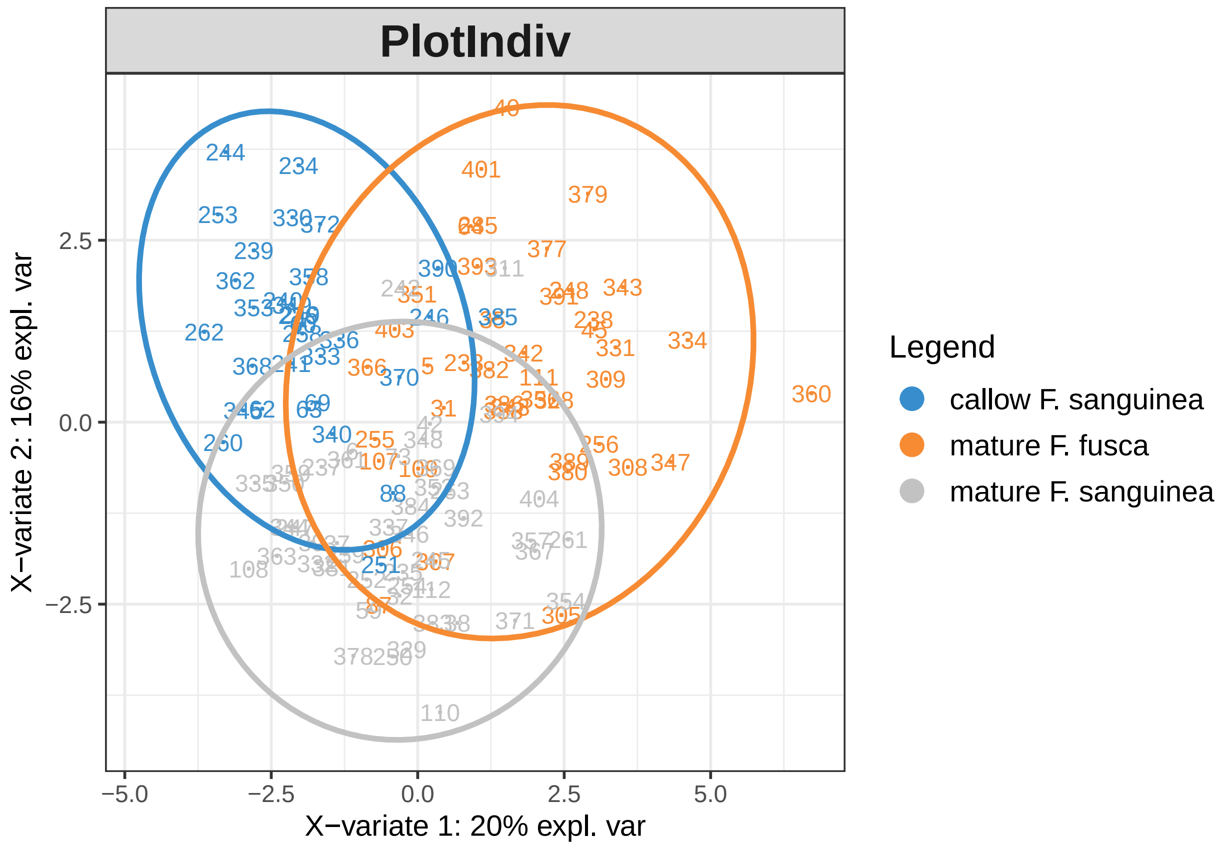


Figure S6 : Projection of the samples in onto two first discriminant analysis components after model tuning.

Table S2 : List of the peaks with the score indicating their importance as a marker of mature *F. fusca*, mature *F. sanguinea*, or callow *F. sanguinea* samples.

Peak ID	Compound	Callow score	Callow marker	Mature score	Mature marker
88	C31ene; 3-MeC30	0.1087104	TRUE	-0.0818747	FALSE
106	C33ene	0.0658706	TRUE	-0.0362677	FALSE
49	13-MeC27	0.0652571	TRUE	-0.0266802	FALSE
37	14-; 10-MeC26 + 3,7,11-triMeC25	0.0560728	TRUE	-0.0455017	FALSE
71	15-; 13-; 11-; 9-; 7-MeC29	0.0512893	TRUE	-0.0065270	FALSE
27	13-; 11-; 9-MeC25	0.0452382	TRUE	-0.0609777	FALSE
59	14-; 12-; 10-MeC28	0.0450478	TRUE	-0.0241388	FALSE
51	5-MeC27	0.0340732	TRUE	0.0078654	FALSE
33	5,13-diMeC25	0.0331561	TRUE	-0.0357471	FALSE
16	12-; 11-; 10-; 9-; 8-MeC24	0.0276485	TRUE	-0.0527224	FALSE
23	4,12-diMeC24	0.0273377	TRUE	-0.0536210	FALSE
43	C27-ene + 6,12-diMeC26	0.0267109	TRUE	0.0257430	FALSE
29	7-MeC25	0.0266817	FALSE	-0.0076892	FALSE
7	C23	0.0243758	FALSE	-0.0329704	FALSE
81	C30	0.0239226	FALSE	-0.0356204	FALSE
41	10,16-; 10,14-diMeC26	0.0212246	FALSE	-0.0158178	FALSE
55	7,11-diMeC27	0.0199886	FALSE	-0.0487542	FALSE
31	11,15-; 9,13-diMeC25	0.0183322	FALSE	-0.0286484	FALSE
58	3,9-; 3,7-diMeC27	0.0173919	FALSE	-0.0305792	FALSE
83	14-; 13-; 12-; 11-MeC30	0.0161514	FALSE	-0.0135349	FALSE
94	11,19-; 11,17-; 11,15-diMeC31	0.0130304	FALSE	0.0493860	TRUE
76	13,17-diMeC29	0.0127919	FALSE	0.0039670	FALSE
12	3-MeC23	0.0112077	FALSE	-0.0155932	FALSE
92	15-; 13-; 11-; 9-MeC31	0.0112060	FALSE	0.0246968	FALSE
36	3,13-diMeC25	0.0109835	FALSE	-0.0089581	FALSE
53	9,17-; 9,15-diMeC27	0.0098728	FALSE	-0.0305881	FALSE
108	x-Me_C32	0.0061566	FALSE	0.0037245	FALSE
56	C28ene	0.0050023	FALSE	0.0311745	TRUE
68	C29	0.0005271	FALSE	-0.0389391	FALSE
101	x-Me_C32	-0.0007951	FALSE	0.0001015	FALSE
8	11-Me-C23 + 9-Me-C23	-0.0022394	FALSE	-0.0063998	FALSE
32	7,11-diMeC25 + 3-MeC25	-0.0031582	FALSE	0.0406657	TRUE
107	x-Me-C32	-0.0080395	FALSE	-0.0144481	FALSE
28	9-MeC25	-0.0099569	FALSE	0.0267150	FALSE
5	C23-ene	-0.0114823	FALSE	-0.0390924	FALSE
30	5-MeC25	-0.0141592	FALSE	0.0564321	TRUE
77	7,11-; 7,13-; 7,15-diMeC29	-0.0150192	FALSE	0.0123666	FALSE
35	C26	-0.0150956	FALSE	-0.0028435	FALSE
14	C24	-0.0154851	FALSE	-0.0162197	FALSE
24	C25	-0.0172402	FALSE	-0.0207323	FALSE
15	3,13-; 3,11-; 3,9-; 3,7-diMeC23	-0.0187089	FALSE	0.0127655	FALSE
54	9,13-diMeC27	-0.0198913	FALSE	0.0362540	TRUE
95	13,19-;13,17-diMeC31	-0.0219664	FALSE	-0.0015095	FALSE
57	C28 + 3,15-diMeC27	-0.0248747	FALSE	0.0186056	FALSE
75	11,17-diMeC29	-0.0260083	FALSE	0.0267627	FALSE

Table S2 : List of the peaks with the score indicating their importance as a marker of mature *F. fusca*, mature *F. sanguinea*, or callow *F. sanguinea* samples. *(continued)*

Peak ID	Compound	Callow score	Callow marker	Mature score	Mature marker
50	7-MeC27	-0.0287298	FALSE	-0.0145294	FALSE
52	11,15-diMeC27	-0.0299602	FALSE	0.0699320	TRUE
109	11,21-; 11,19-diMeC33	-0.0339506	FALSE	0.0446574	TRUE
63	8,12,16-triMeC28	-0.0361913	FALSE	0.0533123	TRUE
9	7-Me-C23	-0.0363853	FALSE	0.0074738	FALSE
85	(di)MeC30 (mix)	-0.0389767	FALSE	0.0621287	TRUE
42	7,x-; 5,x-; 10,14-diMeC26	-0.0413620	FALSE	-0.0302976	FALSE
21	C25-ene	-0.0415401	FALSE	0.0689075	TRUE
90	C31	-0.0587594	FALSE	0.0028683	FALSE
47	C27	-0.0637613	FALSE	0.0327145	TRUE
102	13,17-DiMe_C32 + x,y-DiMe_C32	-0.0867771	FALSE	0.1113618	TRUE

3.1 Verifying the performace of the discrimination model

The performance of the discriminant model was evaluated by examining the accuracy of its predictions in a cross-validation test, measured as the area under the receiver operating characteristic curve (AUROC). This procedure was repeated 10^3 times to account for variance resulting from the random splitting of samples into training and validation sets. In a parallel analysis, the model was trained on data with permuted age status of *F. sanguinea* to assess whether the true assignment of samples influenced model performance. Results from both training approaches were paired, and the difference in AUROC was calculated each for pair. The empirical p-value was defined as the proportion of models in which the AUROC after permutation was equal to or greater than the AUROC based on the original data.

The p-value is calculated as proportion of differences with value equal to or less than zero.

All differences are positive, so the p-value is less than 0.001 since the null distribution consists of 1000 values.

4 Changes of CHCs amount over time

This section contains a series of linear (mixed) models fitted to examine how the proportion of *F. sanguinea* workers in a colony is associated with the amount of CHC, analyzed as a whole or in subsets. Included are summary reports, results of the Shapiro-Wilk test of nornalizty of conditional residuals, as well as diagnostic plots and tests generated with the use of `DHARMa` R package[Hartig, 2024]. The response variable often needed to be transformed to meet model assumptions. Random terms with no variance have been dropped.

In model specification (1|random_factor) denotes random intercept (separate for each level of the random factor) and ((1|random_factor_1:random_factor_2)) denotes random intercepts generated from the combinations of two factors. β_0 and ϵ denote intercept and error term, respectively.

4.1 Change in the total CHCs mass

In this series of models the total CHC mass (normalized by assuming head width of 1.3 mm) was regressed on the proportion of *Formica sanguinea* workers in the colony.

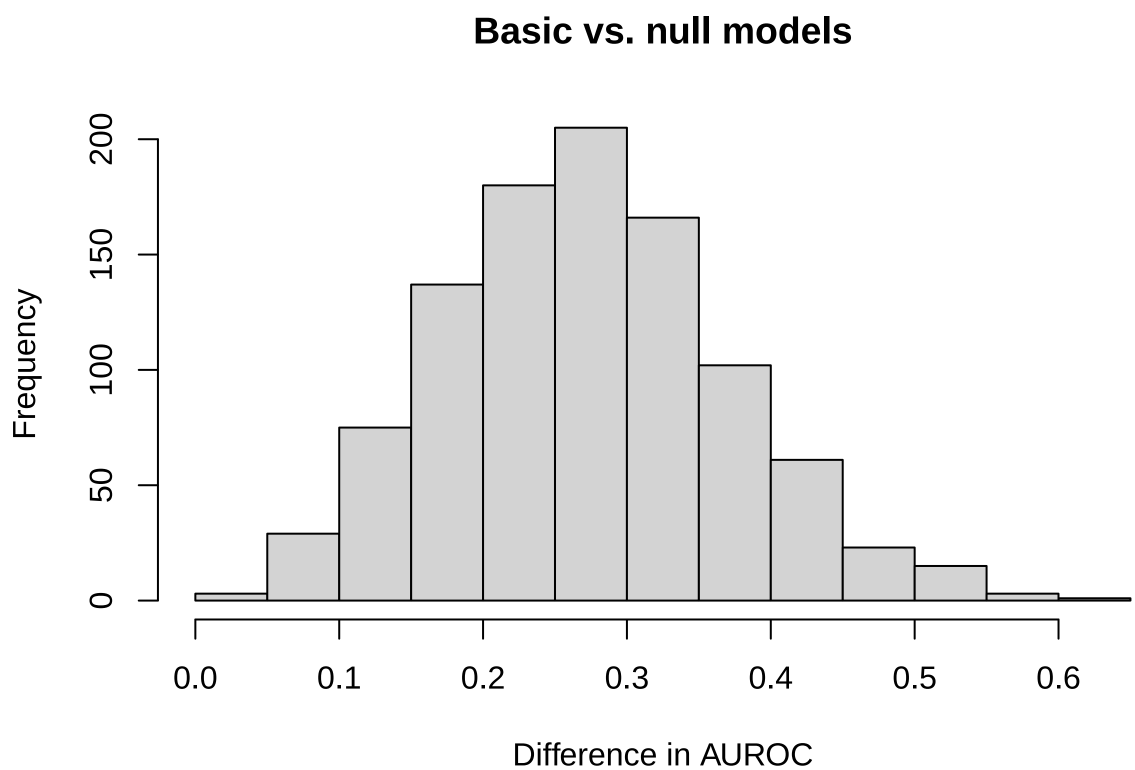


Figure S7 : Distribution of the differences in the performance of the discriminant models trained on true and permuted data. Age status of *F. sanguinea* ants were shuffled in the null variant.

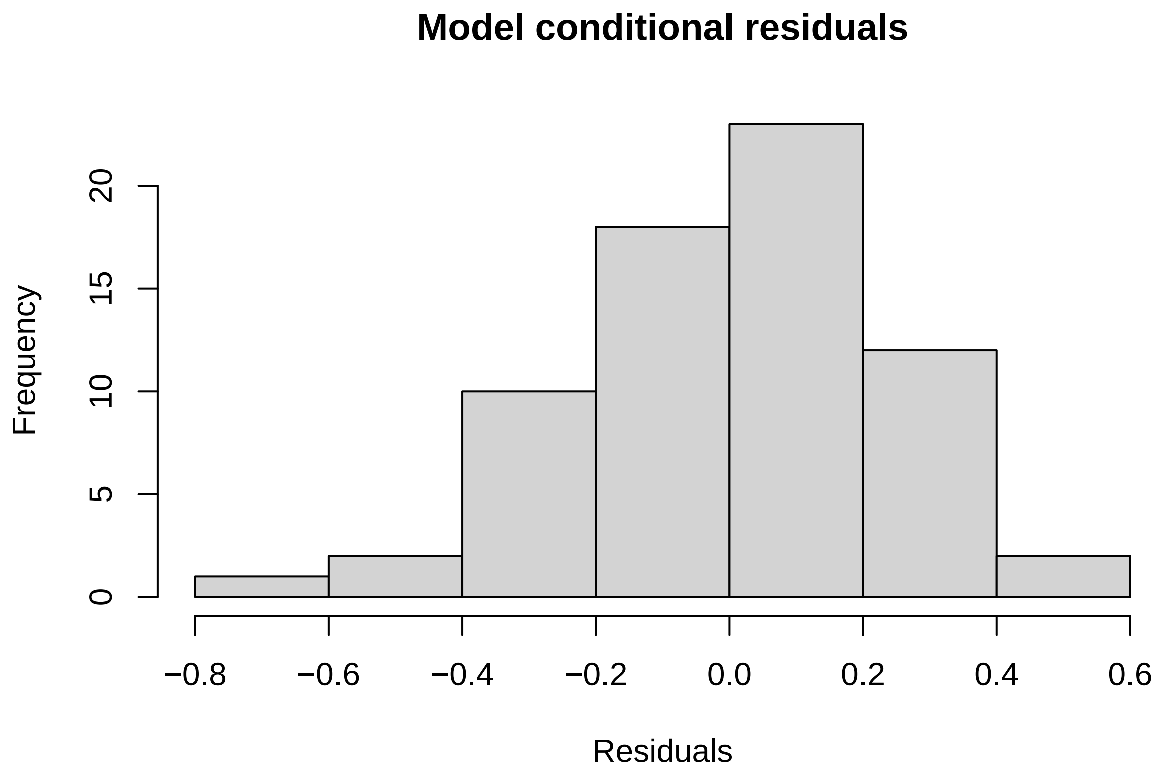
4.1.1 Mature *F. sanguinea*

$$\log(\text{CHC\_mass\_sanguinea}) = \beta_0 + \text{sanguinea\_proportion} + (1|\text{colony\_ID}) + (1|\text{colony\_ID:sampling\_occasion\_ID}) + \epsilon,$$

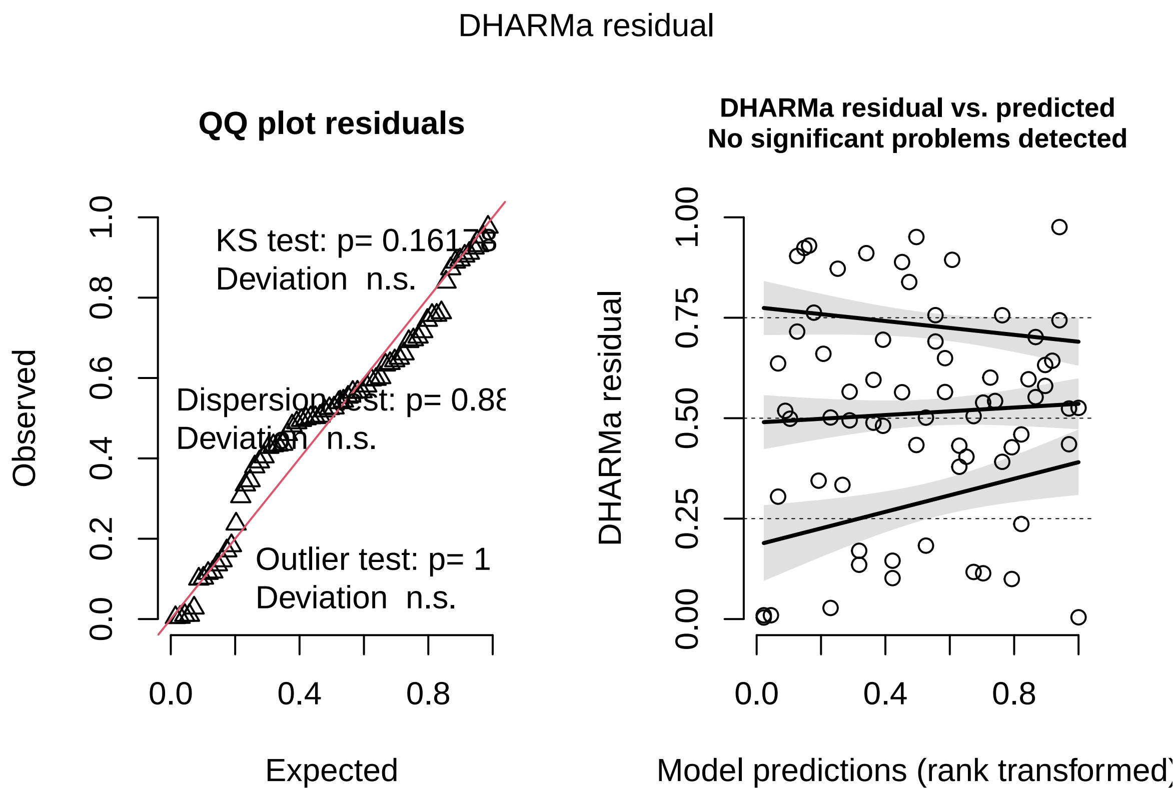
where CHC_mass_sanguinea denotes the total normalized CHC mass on the body of adult *F. sanguinea* worker.

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(log(corrected_mass)) ~ sang_prop + (1 | colony) + (1 | colony:census_date)
##    Data: model_input
##
## REML criterion at convergence: 92.4
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.87332 -0.39129  0.03138  0.53501  1.75167
##
## Random effects:
##  Groups             Name        Variance Std.Dev.
##  colony:census_date (Intercept) 0.13203  0.3634
##  colony             (Intercept) 0.02359  0.1536
##  Residual                       0.10412  0.3227
## Number of obs: 68, groups:  colony:census_date, 44; colony, 16
##
## Fixed effects:
##             Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   0.9145     0.1293 36.2603   7.073 2.53e-08 ***
## sang_prop     0.7815     0.2402 37.2641   3.253  0.00243 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr)
## sang_prop -0.790
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.98549, p-value = 0.6168
```



Model conditional residuals
Frequency
0
5
10
15
20
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
Residuals



4.1.2 *F. fusca*

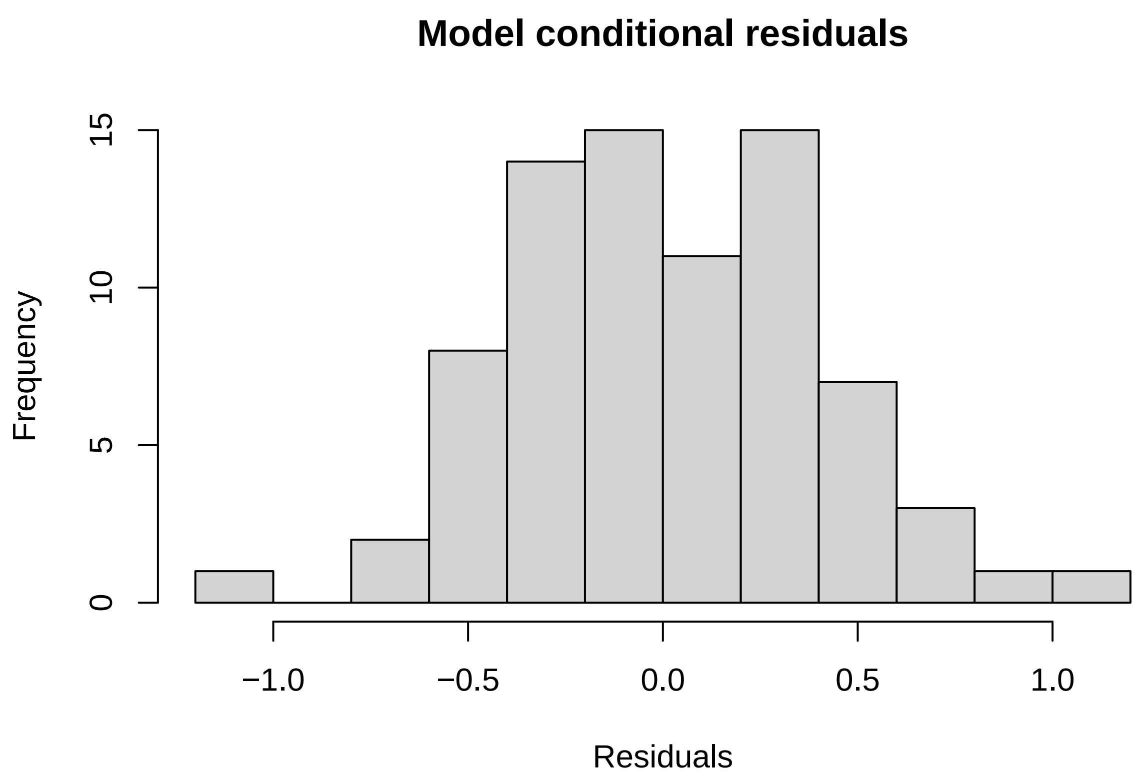
$$\log(\text{CHC\_mass\_fusca}) = \beta_0 + \text{sanguinea\_proportion} + (1|\text{colony\_ID}) + \epsilon,$$

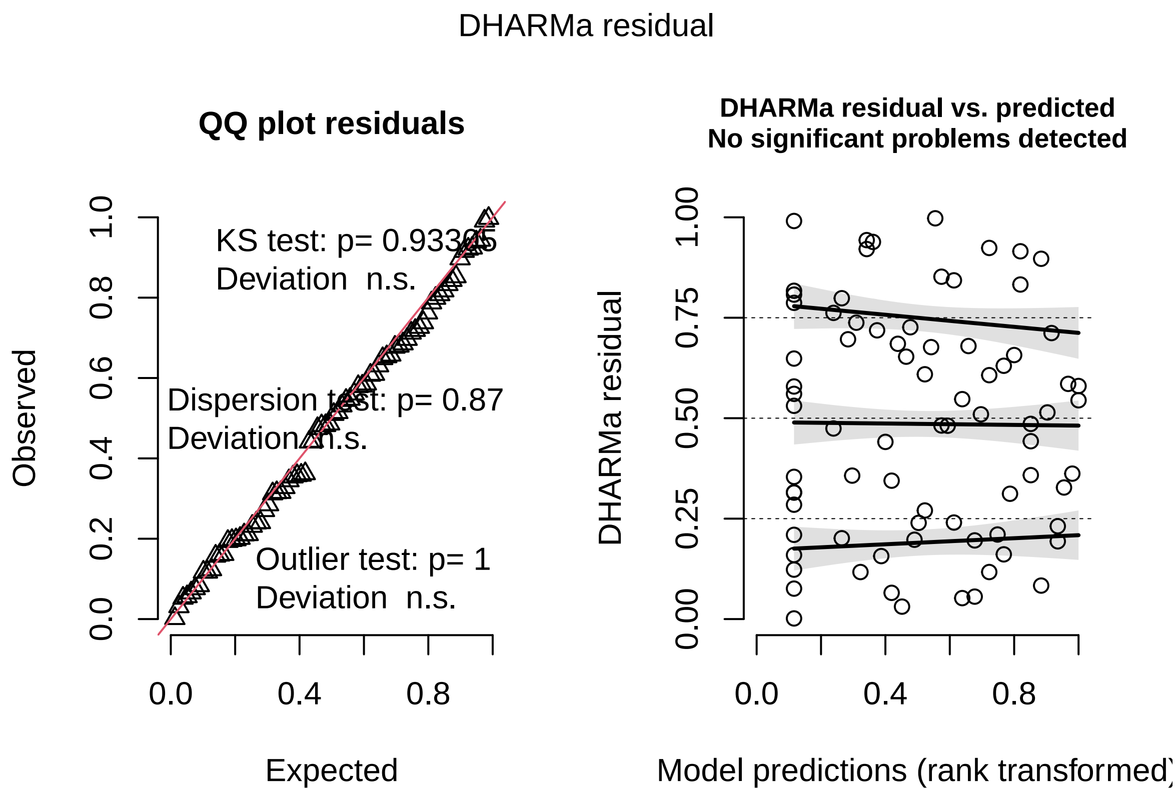
where CHC_mass_fusca denotes the total normalized CHC mass on the body of adult *F. fusca* worker.

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(log(corrected_mass)) ~ sang_prop + (1 | colony)
##    Data: model_input
##
## REML criterion at convergence: 105.4
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.78977 -0.63150 -0.00543  0.63013  2.46223
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  colony   (Intercept) 0.05167  0.2273
##  Residual             0.18054  0.4249
## Number of obs: 78, groups:  colony, 20
##
## Fixed effects:
##             Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)  0.55270    0.08905 31.67460   6.207 6.23e-07 ***
```

```
## sang_prop    0.16713    0.17687 69.98722   0.945    0.348
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr)
## sang_prop -0.592
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.99263, p-value = 0.9379
```





4.1.3 Callow *F. sanguinea*

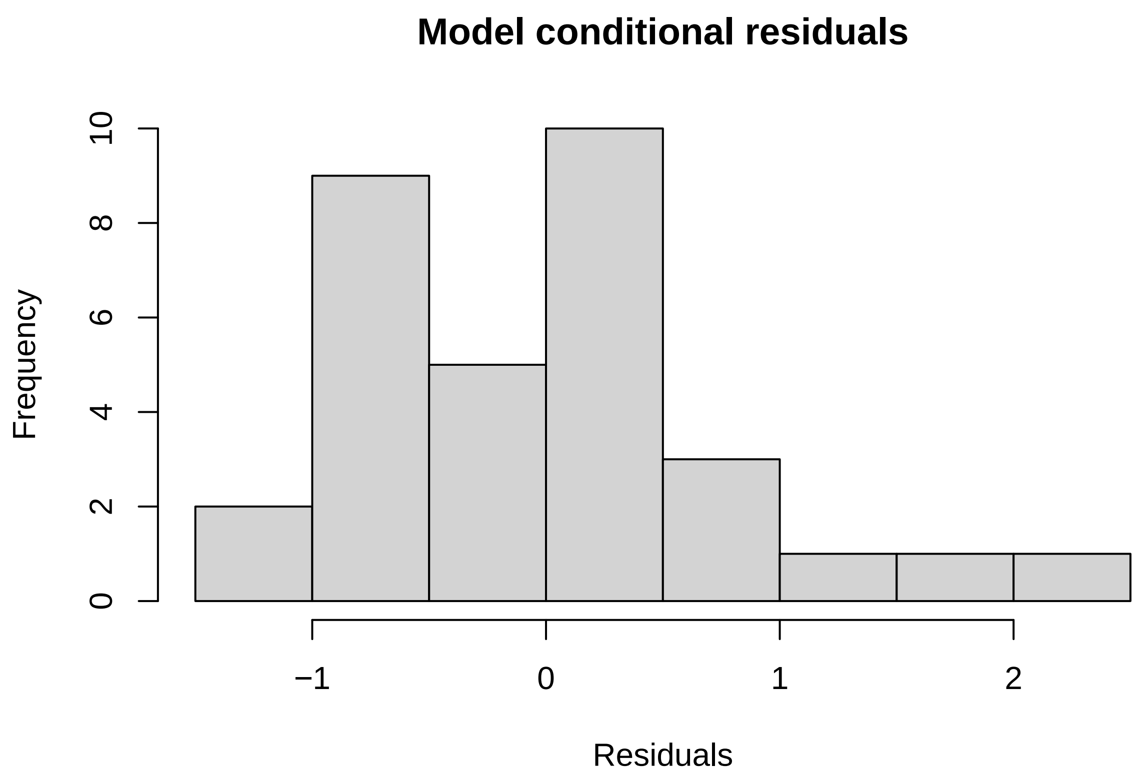
$$\log(\text{CHC\_mass\_callow}) = \beta_0 + \text{sanguinea\_proportion} + (1|\text{colony\_ID}) + (1|\text{colony\_ID:sampling\_occasion\_ID}) + \epsilon,$$

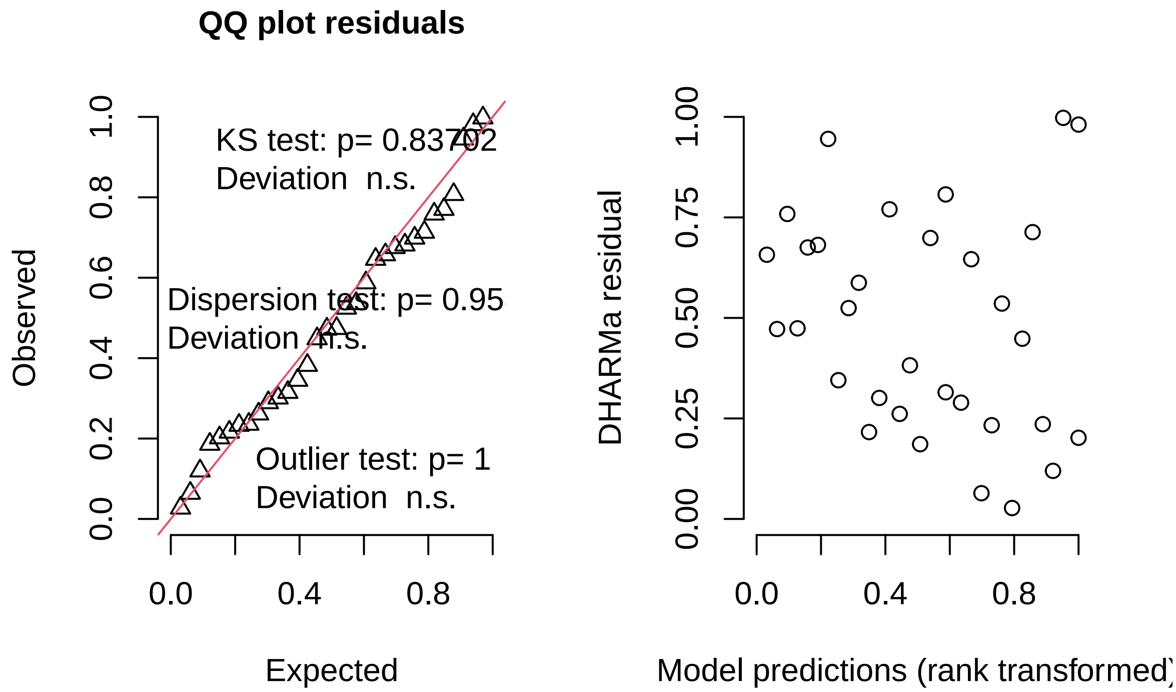
where CHC_mass_callow denotes the total normalized CHC mass on the body of callow *F. sanguinea* worker.

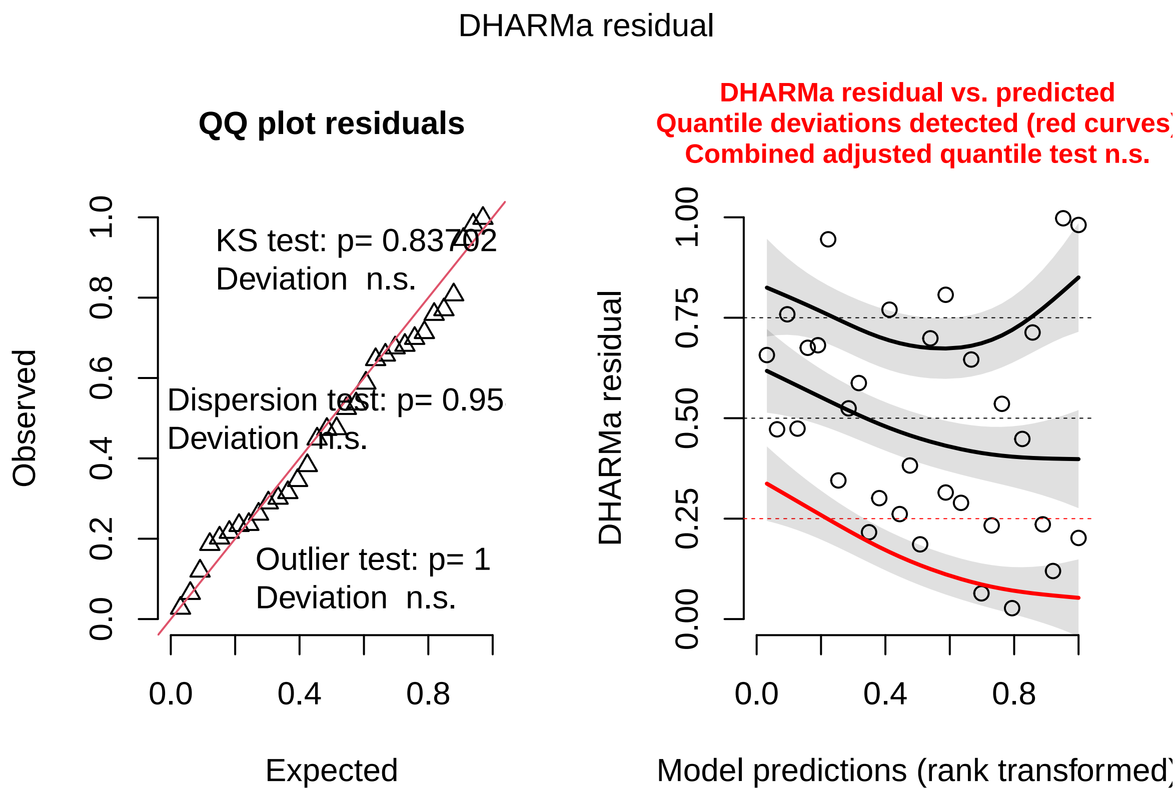
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I((corrected_mass)) ~ sang_prop + (1 | colony)
##    Data: model_input
##
## REML criterion at convergence: 78.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.83633 -0.71584 -0.01041  0.45275  2.72751
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  colony   (Intercept) 0.05088  0.2256
##  Residual             0.63671  0.7979
## Number of obs: 32, groups:  colony, 16
##
## Fixed effects:
##             Estimate Std. Error      df t value Pr(>|t|)
```

```
## (Intercept)   0.6139     0.2630 28.8637   2.335 0.026725 *
## sang_prop     1.9967     0.4866 29.3290   4.103 0.000297 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr)
## sang_prop -0.811
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.94835, p-value = 0.1294
```




QQ plot residuals
KS test: p= 0.83702
Deviation n.s.
Dispersion test: p= 0.95
Deviation n.s.
Outlier test: p= 1
Deviation n.s.
Observed
Expected
DHARMa residual
Model predictions (rank transformed)



4.2 Change of the CHCs associated with *F. sanguinea*

In this section the subset of peaks were used to calculate normalized CHC mass. These were peaks identified as *F. sanguinea* markers.

4.2.1 Mature *F. sanguinea* ants

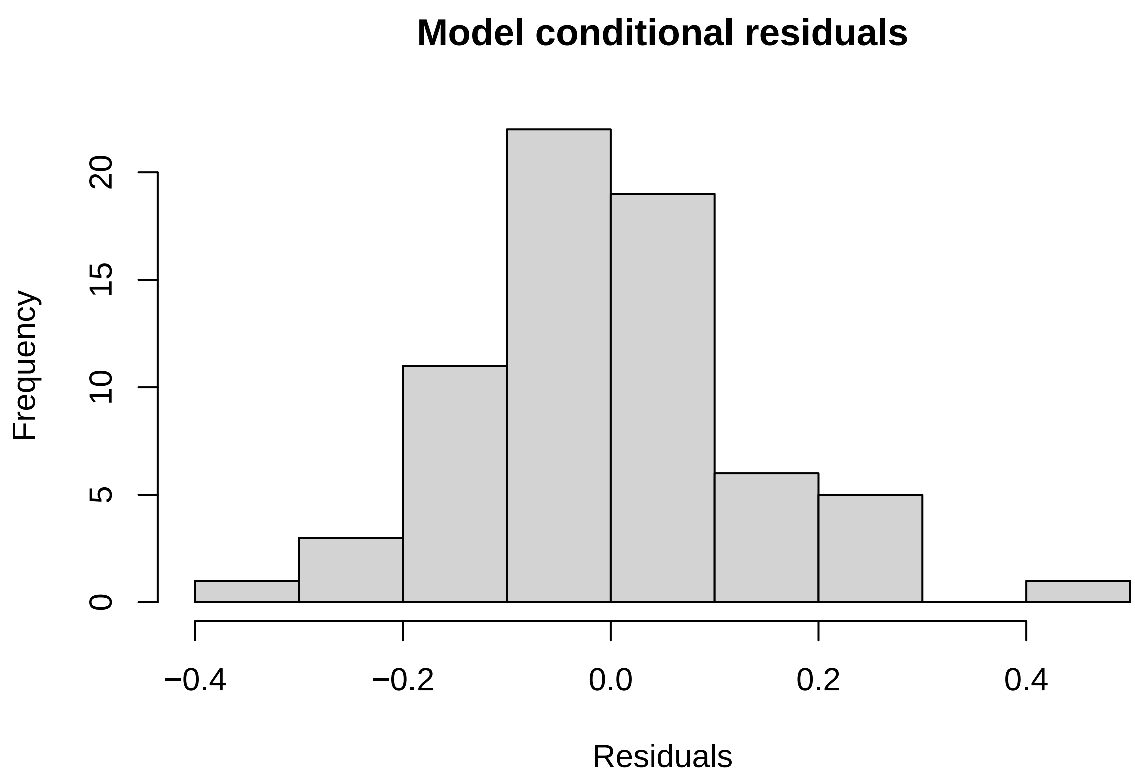
$$\log(\text{CHC\_mass\_mature}) = \beta_0 + \text{sanguinea\_proportion} + (1|\text{colony\_ID}) + (1|\text{colony\_ID:sampling\_occasion\_ID}) + \epsilon,$$

where CHC_mass_mature denotes the normalized mass of *F. sanguinea* markers on the body of adult *F. sanguinea* worker.

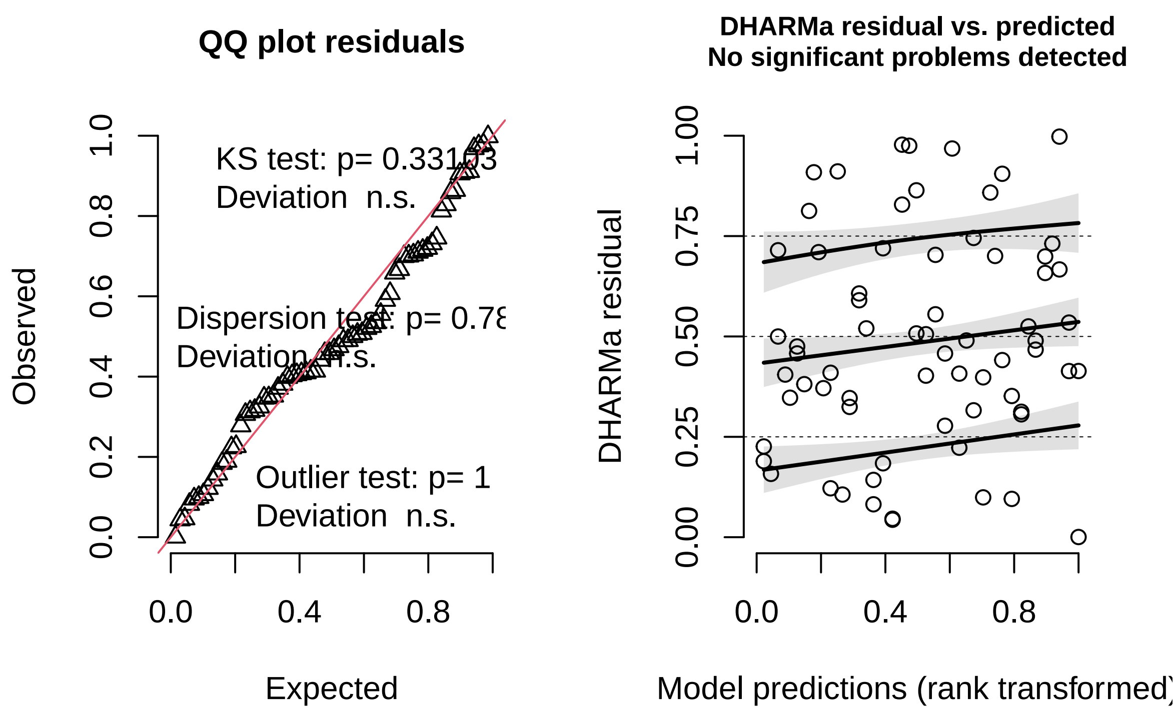
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(log(corrected_mass + 1)) ~ sang_prop + (1 | colony:census_date) +      (1 | colony)
##    Data: model_input
##
## REML criterion at convergence: 18.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.98276 -0.35520 -0.07171  0.39288  2.46714
##
## Random effects:
##  Groups             Name        Variance Std.Dev.
##  colony:census_date (Intercept) 0.05192  0.2279
```

```
##  colony             (Intercept) 0.00000  0.0000
##  Residual                       0.03265  0.1807
## Number of obs: 68, groups:  colony:census_date, 44; colony, 16
##
## Fixed effects:
##             Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)  0.22604    0.07252 40.08622   3.117  0.00337 **
## sang_prop    1.17085    0.14090 39.52090   8.310 3.32e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr)
## sang_prop -0.823
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.97833, p-value = 0.2827
```



DHARMa residual



4.2.2 *F. fusca* ants

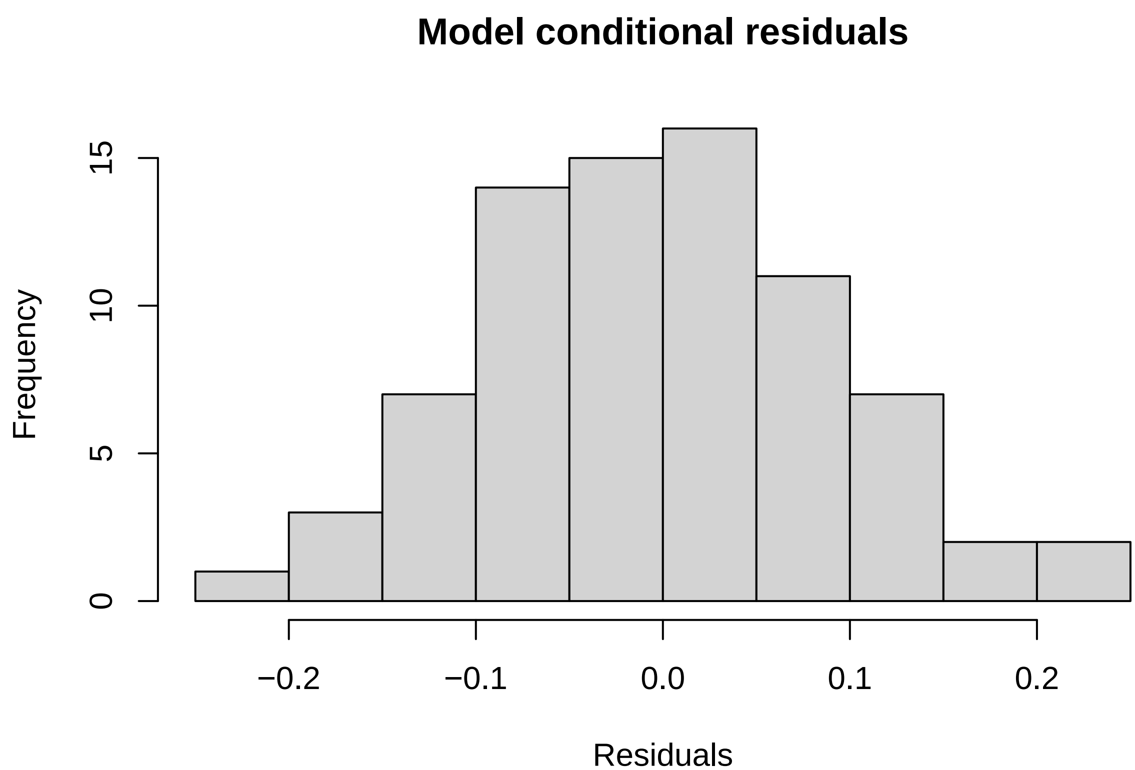
$\sqrt{(\text{CHC\_mass\_fusca})} = \beta_0 + \text{sanguinea\_proportion} + (1|\text{colony\_ID}) + (1|\text{colony\_ID:sampling\_occasion\_ID}) + \epsilon,$

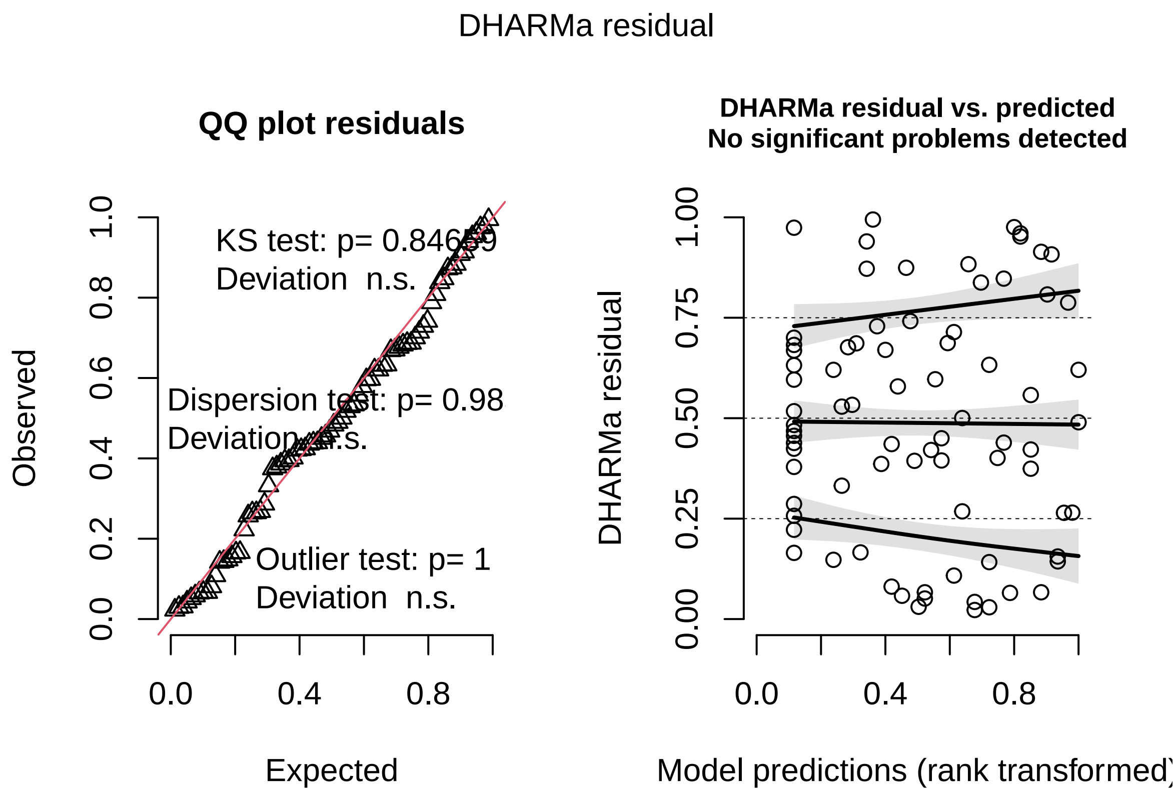
where CHC_mass_fusca denotes the normalized mass of *F. sanguinea* markers on the body of adult *F. sanguinea* worker.

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(sqrt(corrected_mass)) ~ sang_prop + (1 | colony:census_date)
##    Data: model_input
##
## REML criterion at convergence: -69.4
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.95510 -0.51415 -0.02971  0.47328  1.91504
##
## Random effects:
##  Groups             Name        Variance Std.Dev.
##  colony:census_date (Intercept) 0.01008  0.1004
##  Residual                       0.01316  0.1147
## Number of obs: 78, groups:  colony:census_date, 55
##
## Fixed effects:
##             Estimate Std. Error       df t value Pr(>|t|)
```

```
## (Intercept)  0.39488    0.02786 53.89308  14.175  < 2e-16 ***
## sang_prop    0.53379    0.06586 51.71118   8.105 9.03e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr)
## sang_prop -0.728
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.99409, p-value = 0.9782
```





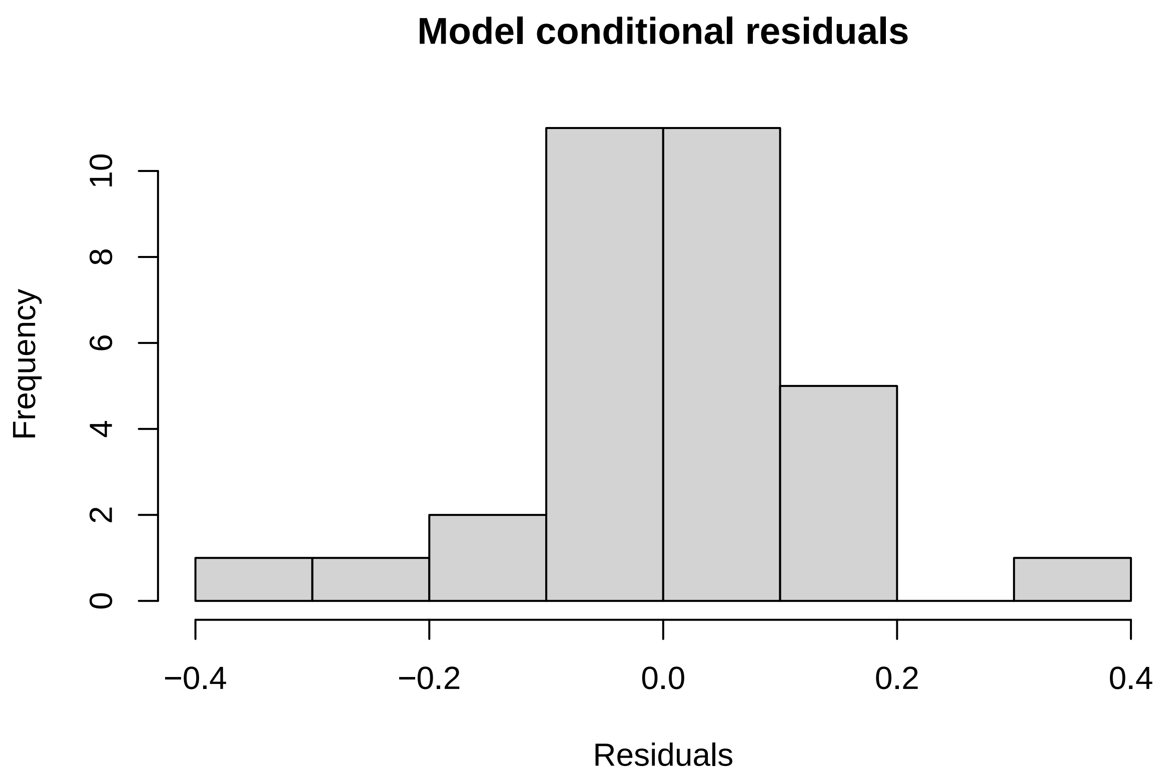
4.2.3 Callow *F. sanguinea* ants

$$\sqrt[3]{(\text{CHC\_mass\_mature})} = \beta_0 + \text{sanguinea\_proportion} + \epsilon,$$

where CHC_mass_mature denotes the normalized mass of *F. sanguinea* markers on the body of callow *F. sanguinea* worker.

```
##
## Call:
## lm(formula = I((corrected_mass)^(1/3)) ~ sang_prop, data = model_input)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.38219 -0.06276  0.02196  0.07257  0.32302
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.29654    0.04252   6.975 9.50e-08 ***
## sang_prop    0.64225    0.07993   8.035 5.72e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1351 on 30 degrees of freedom
## Multiple R-squared:  0.6828, Adjusted R-squared:  0.6722
## F-statistic: 64.57 on 1 and 30 DF,  p-value: 5.716e-09
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.96934, p-value = 0.4815
```



4.3 Change of the CHCs associated with *F. fusca*

Similarly to the analysis with the use of markers of *F. sanguinea* workers, the procedure was repeated using the mass of *F. fusca* markers as a response variable.

4.3.1 Mature *F. sanguinea* ants

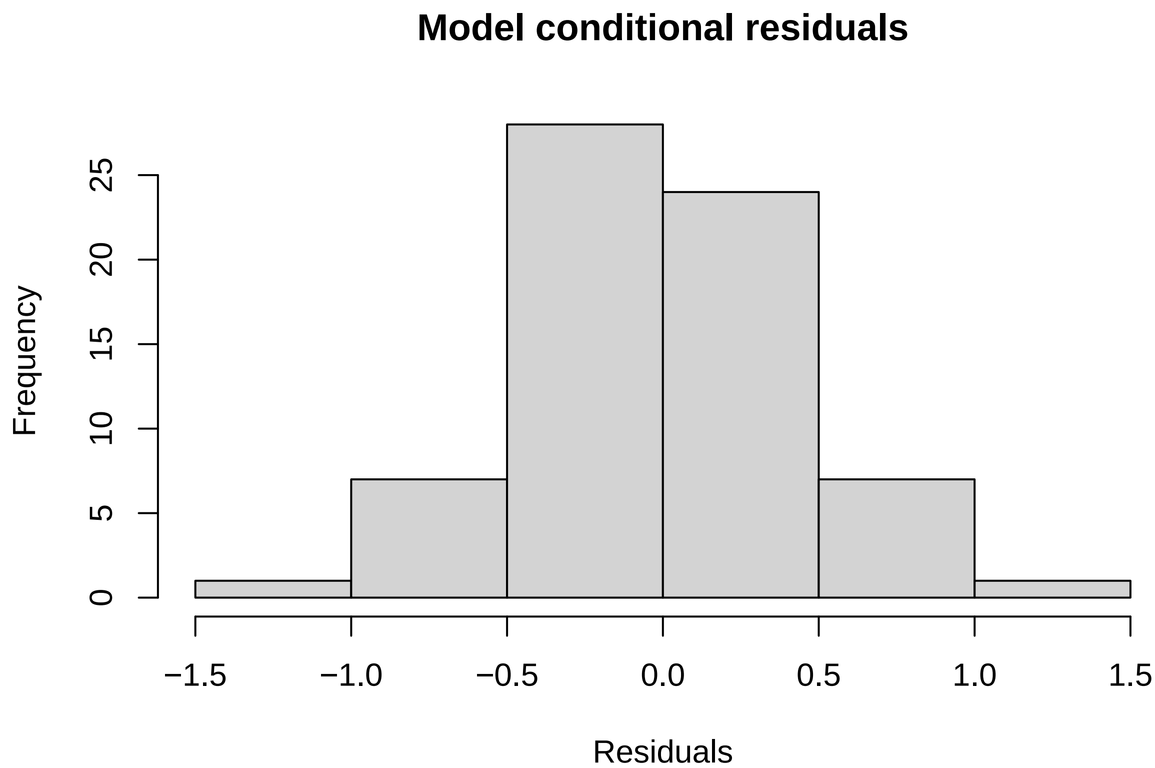
$$\log(\text{CHC\_mass\_mature}) = \beta_0 + \text{sanguinea\_proportion} + (1|\text{colony\_ID}) + (1|\text{colony\_ID:sampling\_occasion\_ID}) + \epsilon,$$

where CHC_mass_mature denotes the normalized mass of *F. fusca* markers on the body of adult *F. sanguinea* worker.

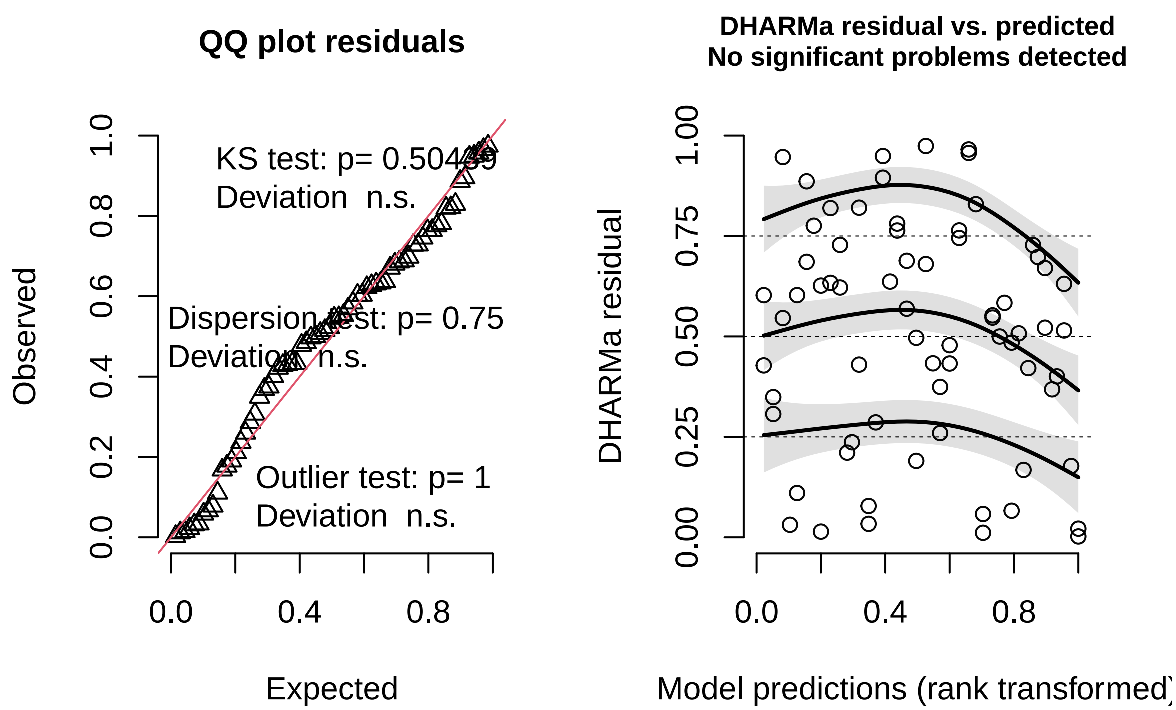
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(log(corrected_mass)) ~ sang_prop + (1 | colony:census_date) +      (1 | colony)
##    Data: model_input
##
## REML criterion at convergence: 154.6
##
```

```
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.32675 -0.47439 -0.04918  0.42137  2.01162
##
## Random effects:
##  Groups             Name        Variance Std.Dev.
##  colony:census_date (Intercept) 0.1184   0.3442
##  colony             (Intercept) 0.3702   0.6084
##  Residual                       0.3039   0.5513
## Number of obs: 68, groups:  colony:census_date, 44; colony, 16
##
## Fixed effects:
##             Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  -0.5686     0.2268 31.9619  -2.507   0.0174 *
## sang_prop    -1.8817     0.3262 30.8622  -5.769 2.41e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr)
## sang_prop -0.626
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.98815, p-value = 0.7693
```


Model conditional residuals
Frequency
0
5
10
15
20
25
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Residuals

DHARMa residual



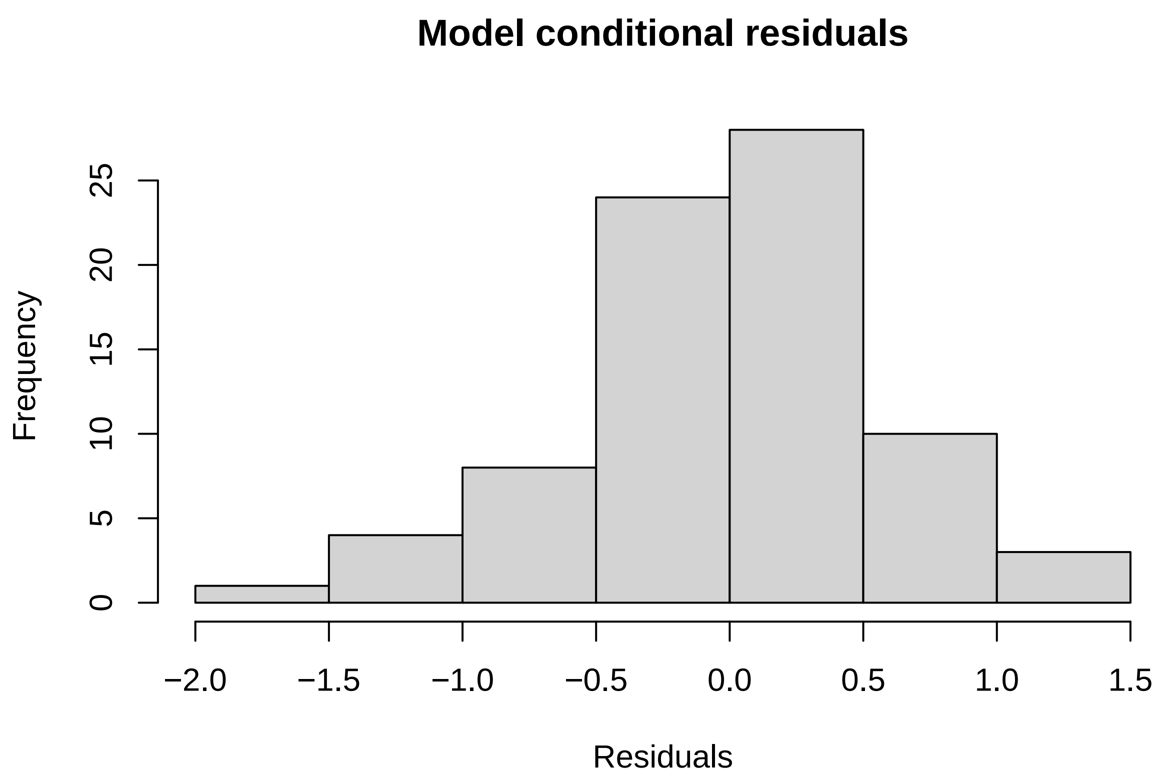
4.3.2 *F. fusca* ants

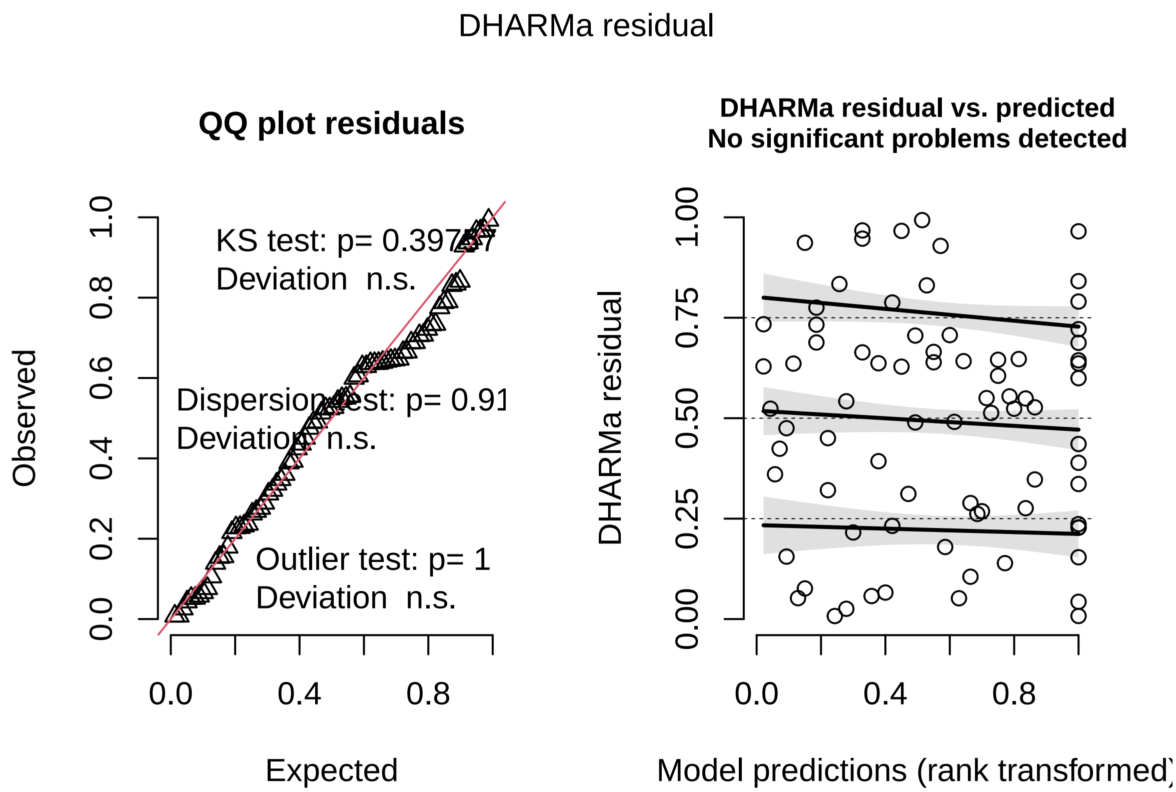
$\log(\text{CHC\_mass\_mature}{+}0.01) = \beta_0{+}\text{sanguinea\_proportion}{+}(1|\text{colony\_ID}){+}(1|\text{colony\_ID:sampling\_occasion\_ID}){+}\epsilon$,
where CHC_mass_mature denotes the normalized mass of *F. fusca* markers on the body of *F. fusca* worker.

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(log(corrected_mass + 0.01)) ~ sang_prop + (1 | colony) + (1 |      colony:census_date)
##    Data: model_input
##
## REML criterion at convergence: 174.2
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.4825 -0.5952  0.0330  0.6208  2.2609
##
## Random effects:
##  Groups             Name        Variance Std.Dev.
##  colony:census_date (Intercept) 0.0000   0.0000
##  colony             (Intercept) 0.2241   0.4734
##  Residual                       0.4066   0.6377
## Number of obs: 78, groups:  colony:census_date, 55; colony, 20
##
## Fixed effects:
##             Estimate Std. Error      df t value Pr(>|t|)
```

```
## (Intercept)  -1.0945     0.1532 30.5988  -7.144 5.37e-08 ***
## sang_prop    -1.4650     0.2716 67.8691  -5.395 9.40e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr)
## sang_prop -0.516
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.98927, p-value = 0.7624
```





4.3.3 Callow *F. sanguinea* ants

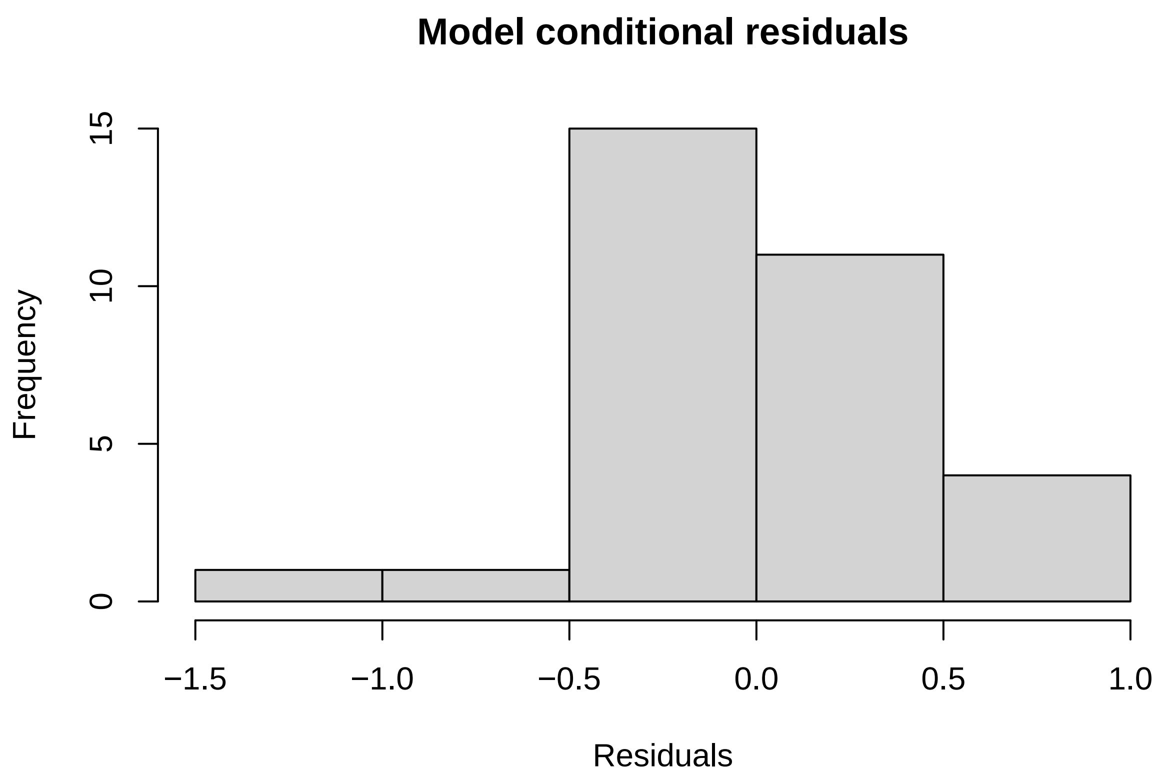
$$\log(\text{CHC\_mass\_mature}) = \beta_0 + \text{sanguinea\_proportion} + (1|\text{colony\_ID}) + \epsilon,$$

where log(CHC_mass_mature) denotes the normalized mass of *F. fusca* markers on the body of callow *F. sanguinea* worker.

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(log(corrected_mass)) ~ sang_prop + (1 | colony)
##    Data: model_input
##
## REML criterion at convergence: 64.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.29425 -0.57286 -0.03774  0.61675  1.83684
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  colony   (Intercept) 0.2338   0.4835
##  Residual             0.2733   0.5228
## Number of obs: 32, groups:  colony, 16
##
## Fixed effects:
##             Estimate Std. Error      df t value Pr(>|t|)
```

```
## (Intercept)  -1.9565     0.2187 28.3341  -8.944 9.58e-10 ***
## sang_prop    -0.1866     0.3551 20.4211  -0.525    0.605
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr)
## sang_prop -0.700
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.97969, p-value = 0.7905
```

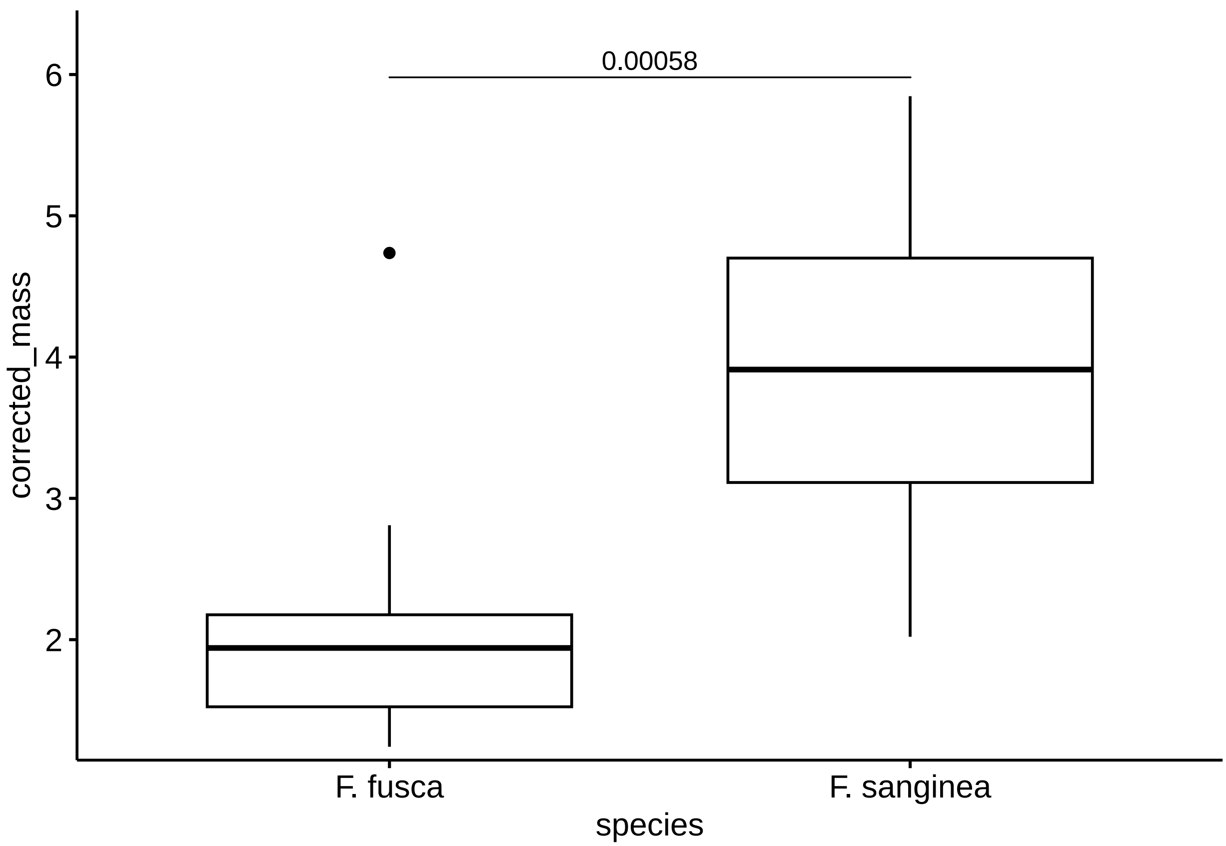


5 Comparison of the amount and proportions of CHC between species and age categories

This section presents the results of Wilcoxon tests comparing features of CHC profiles between callow and mature *F. sanguinea* ants as well as *F. fusca* slaves. Samples from the same colony were averaged before calculation of the final statistics to account for their non-independence.

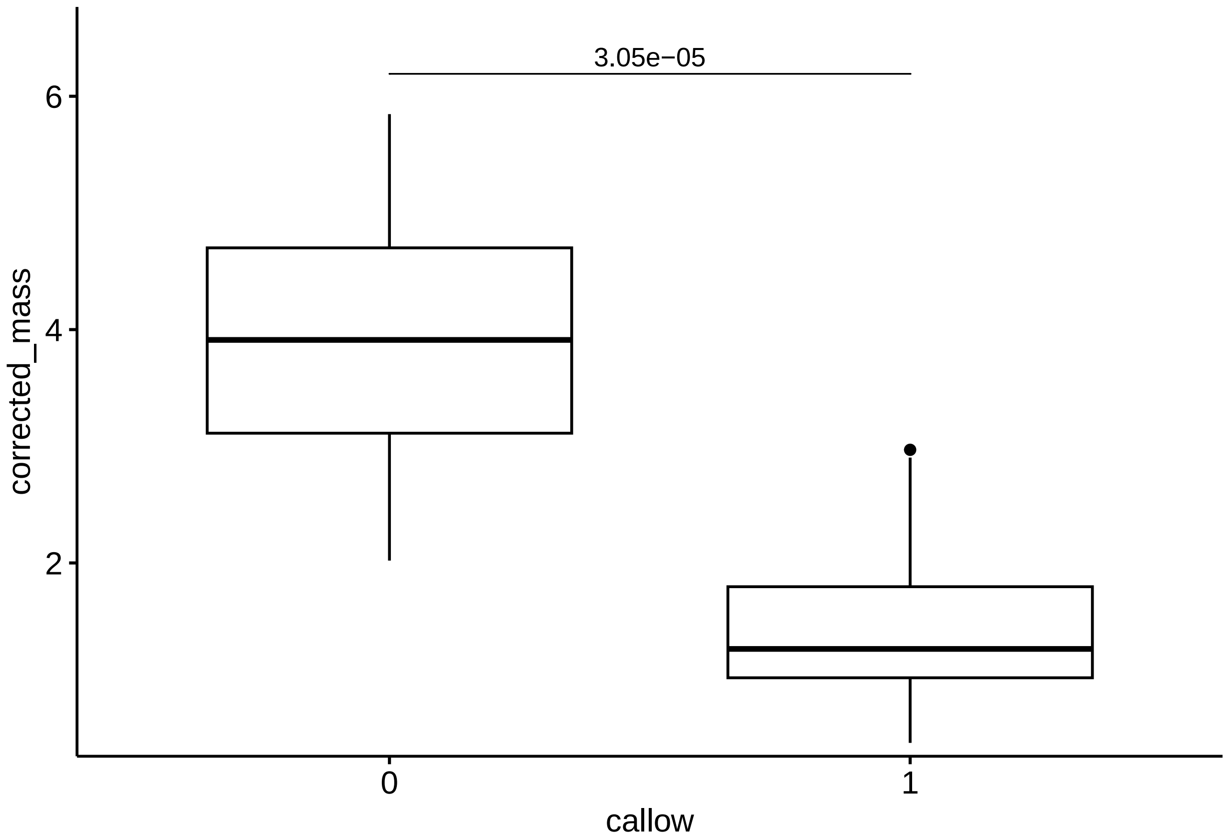
5.1 Difference in the total CHC amount between *F. sanguinea* and *F. fusca*

```
## # A tibble: 1 x 7
##   .y.            group1   group2         n1    n2 statistic       p
## * <chr>          <chr>    <chr>       <int> <int>     <dbl>   <dbl>
## 1 corrected_mass F. fusca F. sanginea    16    16         7 0.00058
```



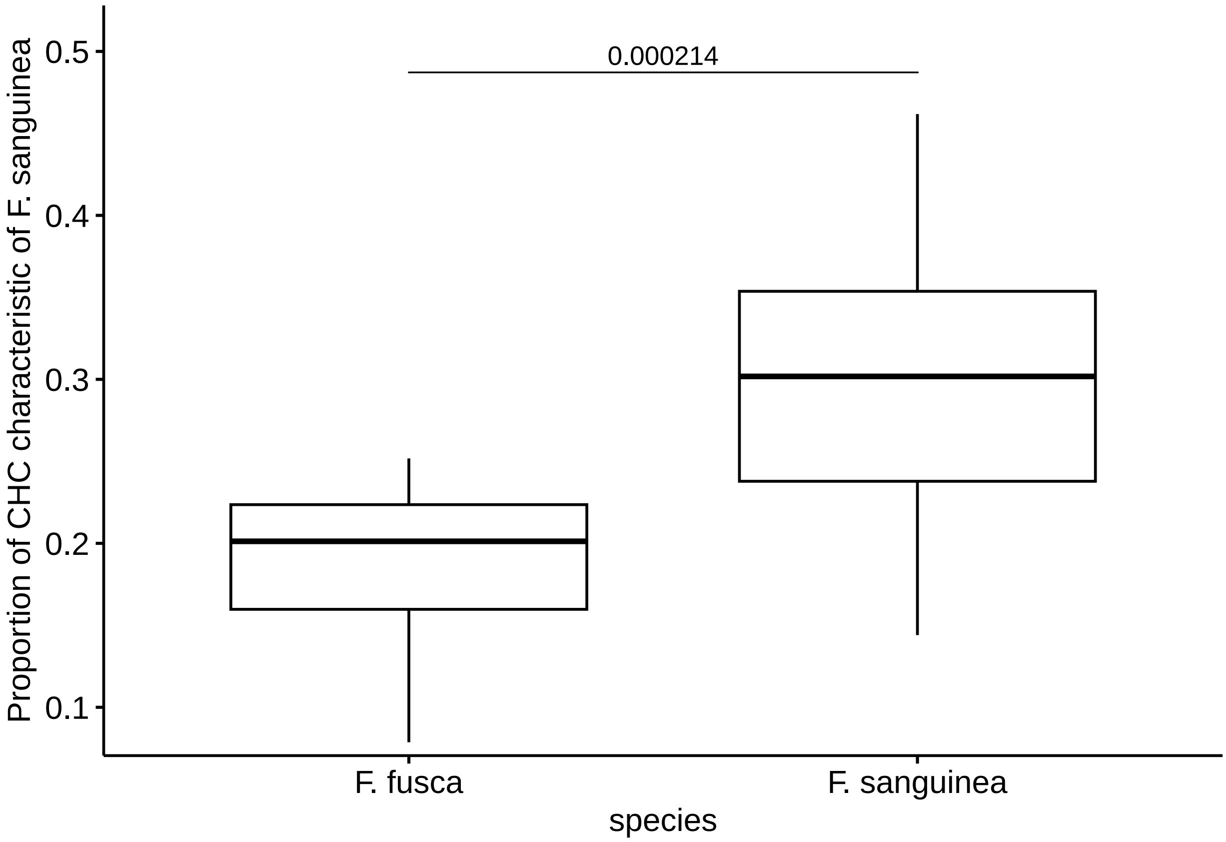
5.2 Difference in the total CHC amount between callow and mature *F. sanguinea*

```
## # A tibble: 1 x 7
##   .y.            group1 group2    n1    n2 statistic         p
## * <chr>          <chr>  <chr>  <int> <int>     <dbl>     <dbl>
## 1 corrected_mass 0      1         16    16       136 0.0000305
```

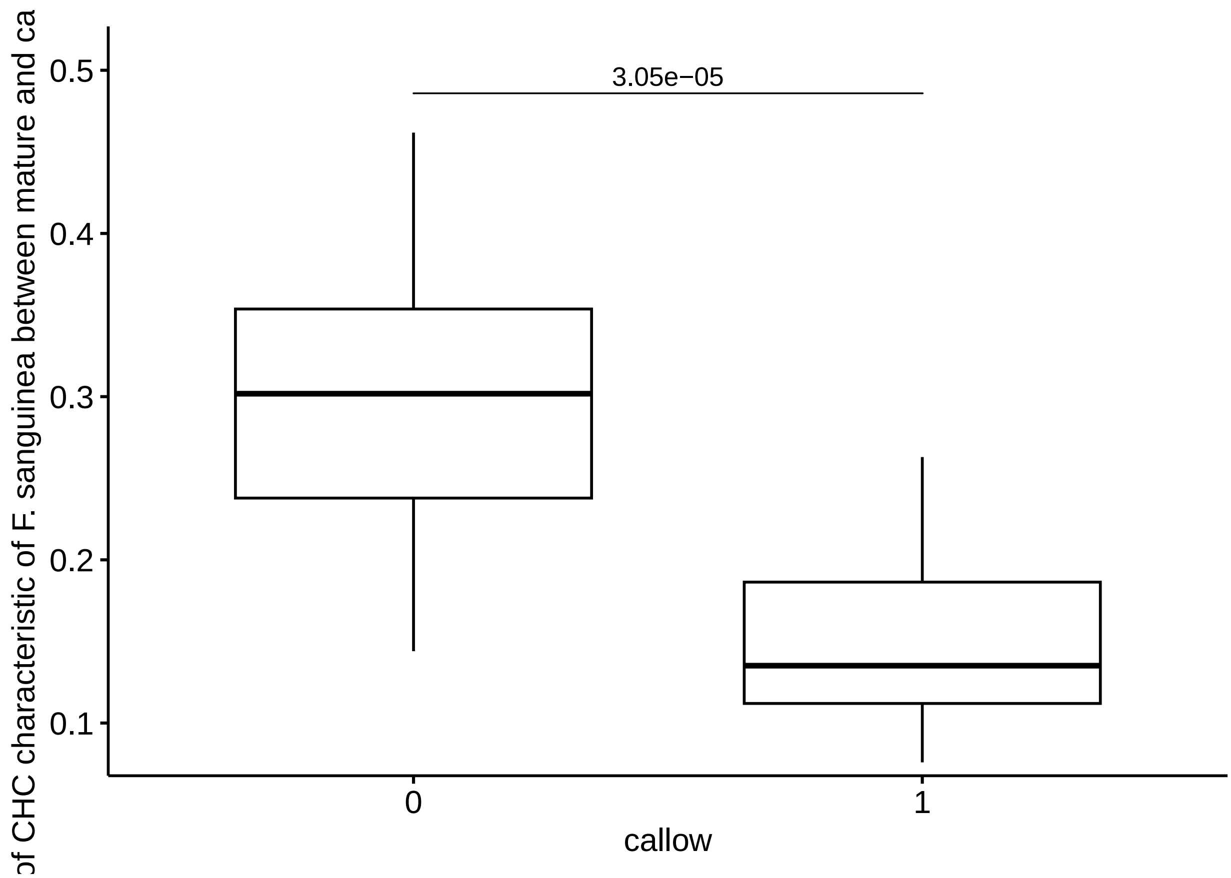


3.05e−05
6
4
2
corrected_mass
0
1
callow

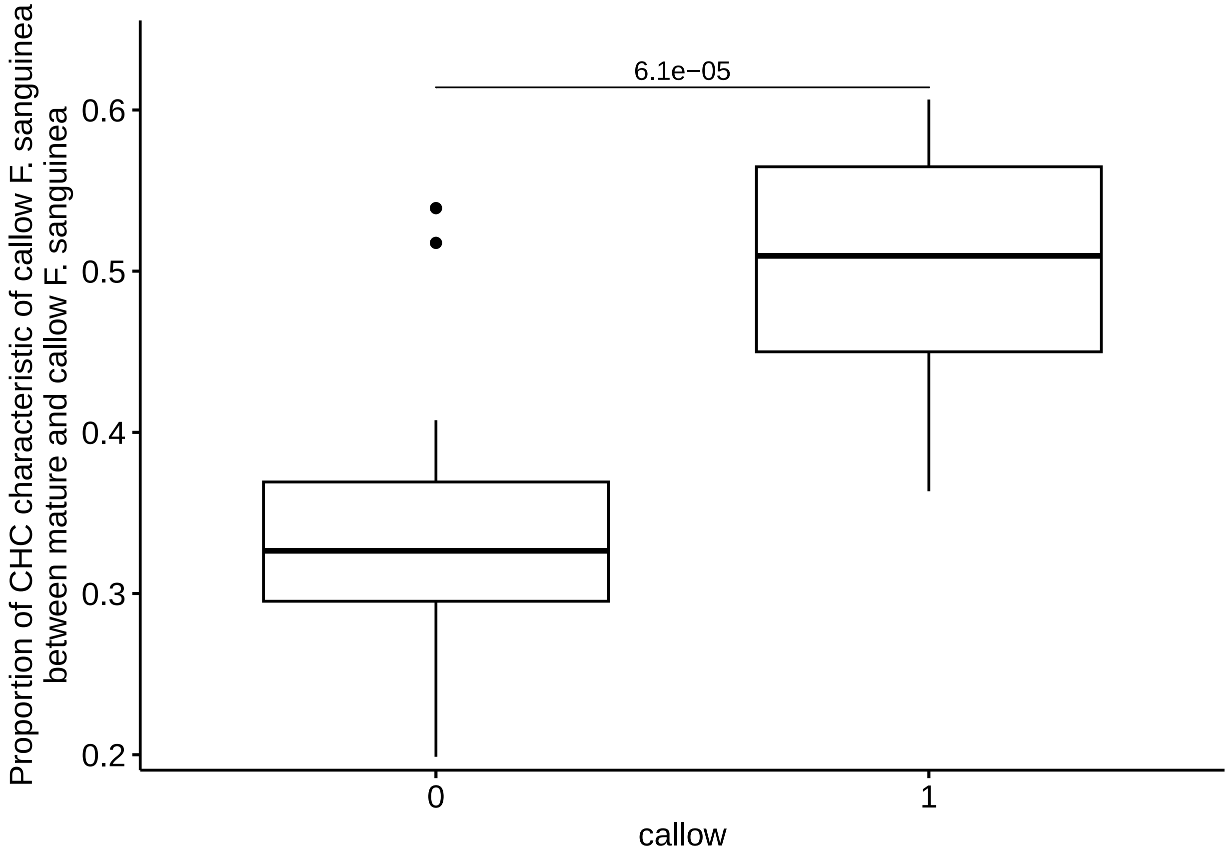
5.3 Difference in the proportion of CHC characteristic of *F. sanguinea* between mature *F. sanguinea* and *F. fusca*



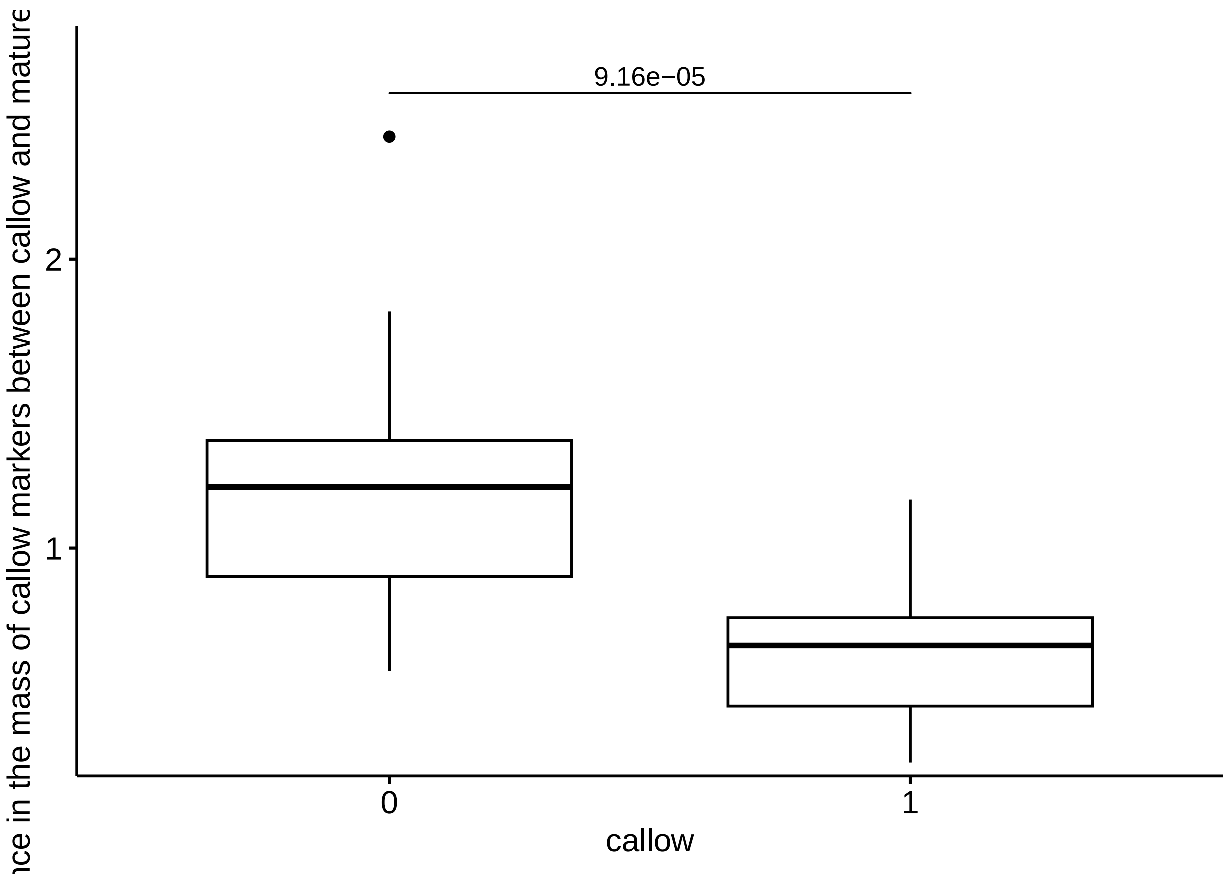
5.4 Difference in the proportion of CHC characteristic of *F. sanguinea* between mature and callow *F. sanguinea*



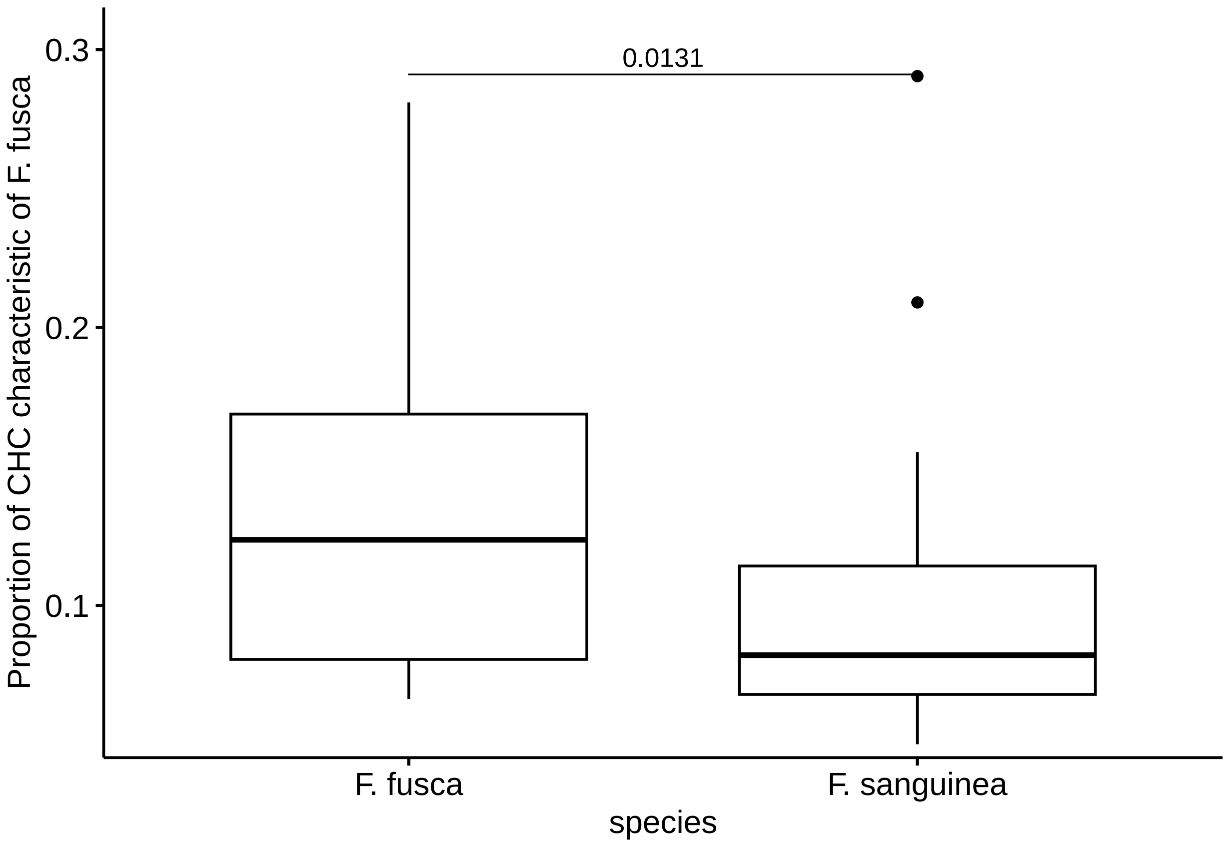
5.5 Difference in the proportion of CHC characteristic of callow *F. sanguinea* between mature and callow *F. sanguinea*



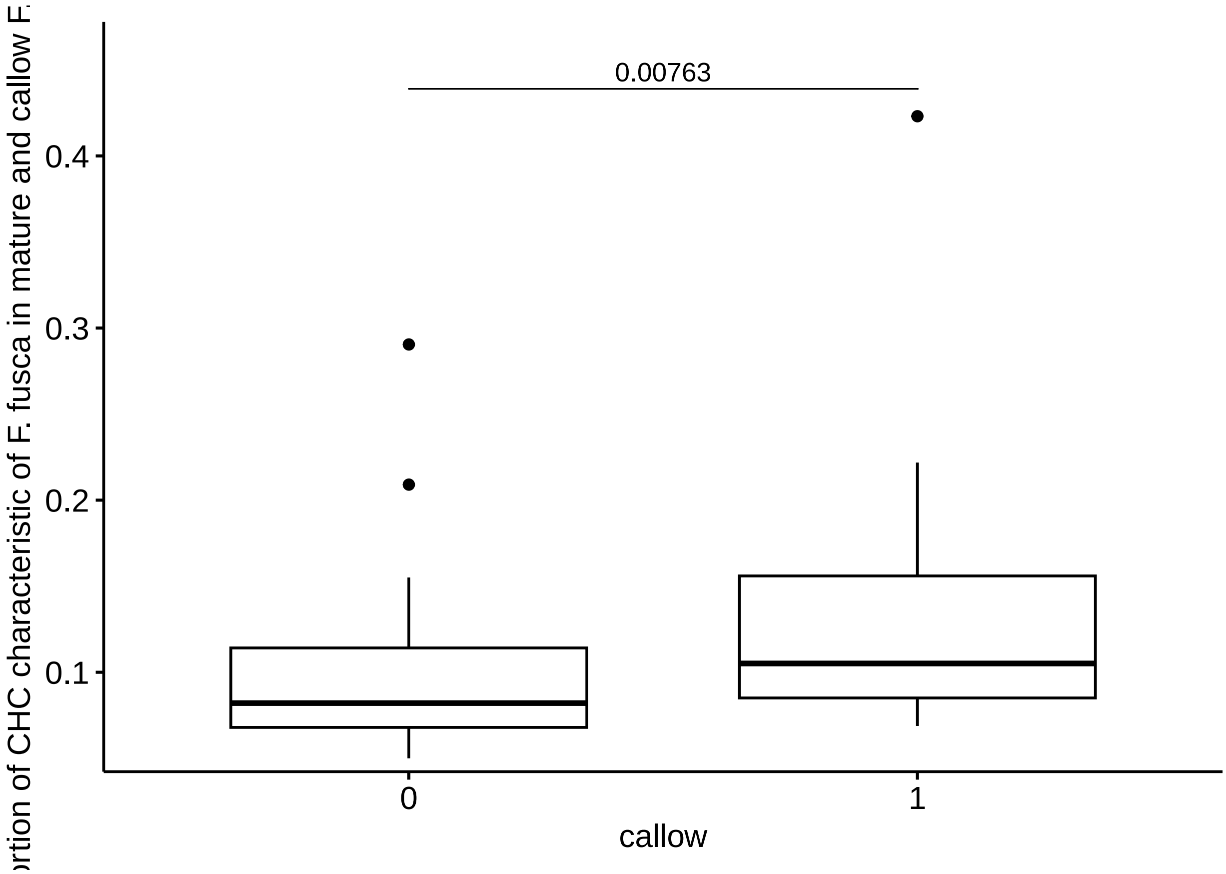
5.6 Difference in the amount of CHC characteristic of callow *F. sanguinea* between mature and callow *F. sanguinea*



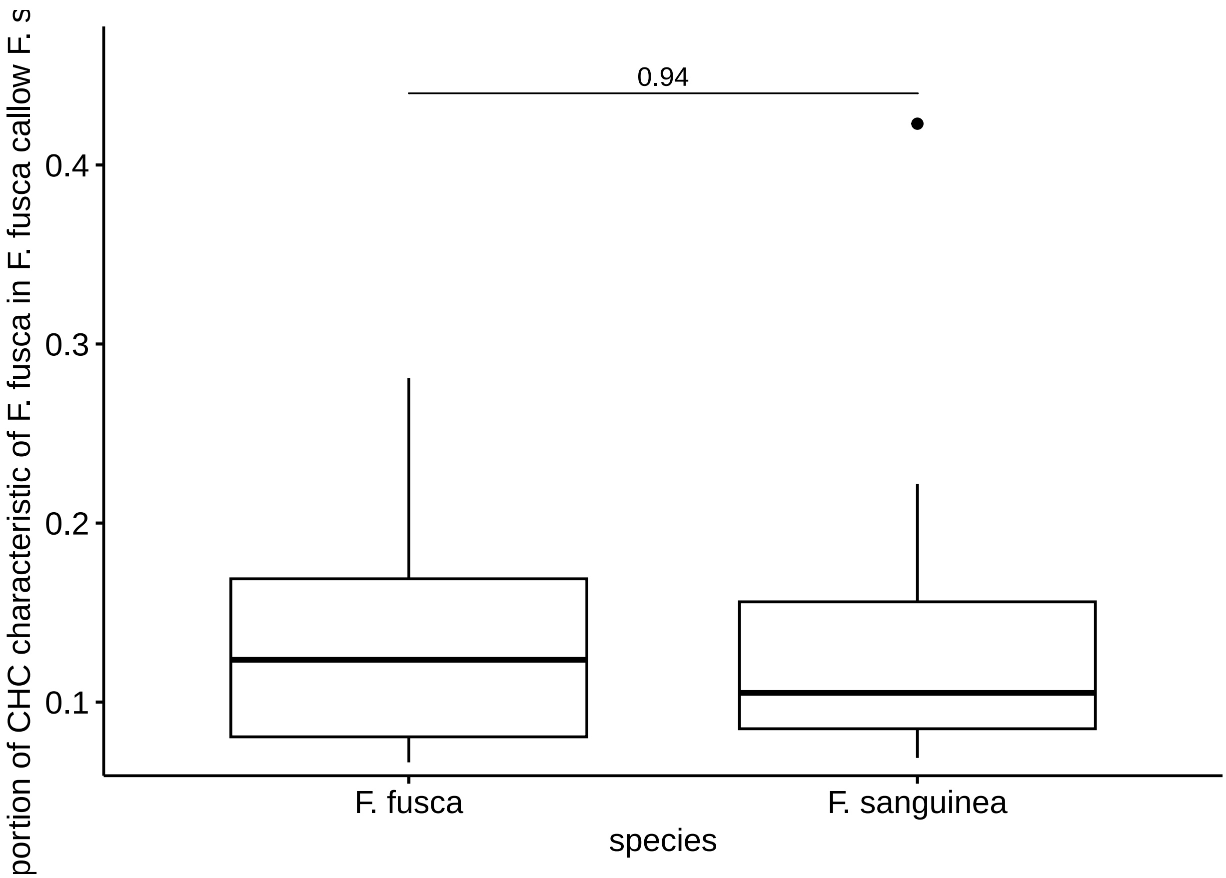
5.7 Difference in the proportion of CHC characteristic of *F. fusca* between mature *F. sanguinea* and *F. fusca*



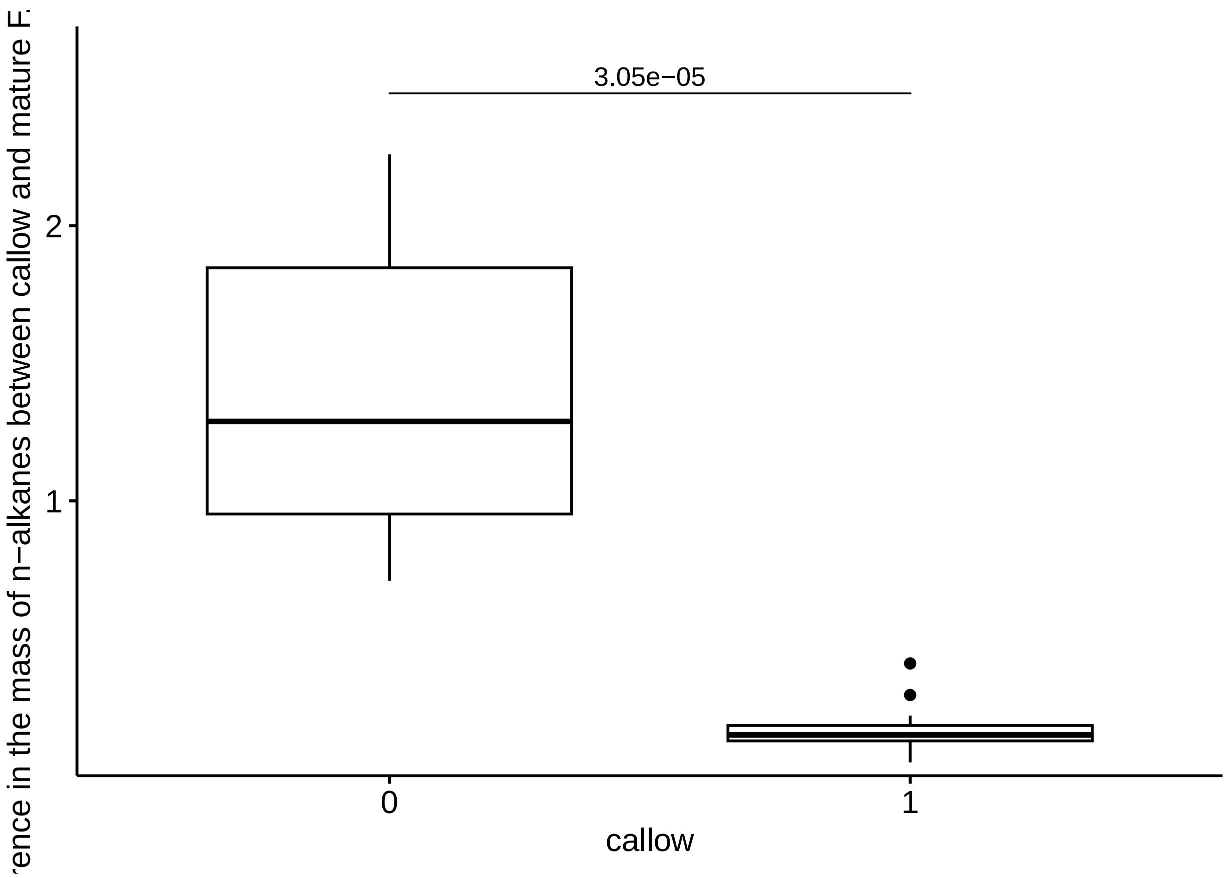
5.8 Difference in the proportion of CHC characteristic of *F. fusca* between mature and callow *F. sanguinea*



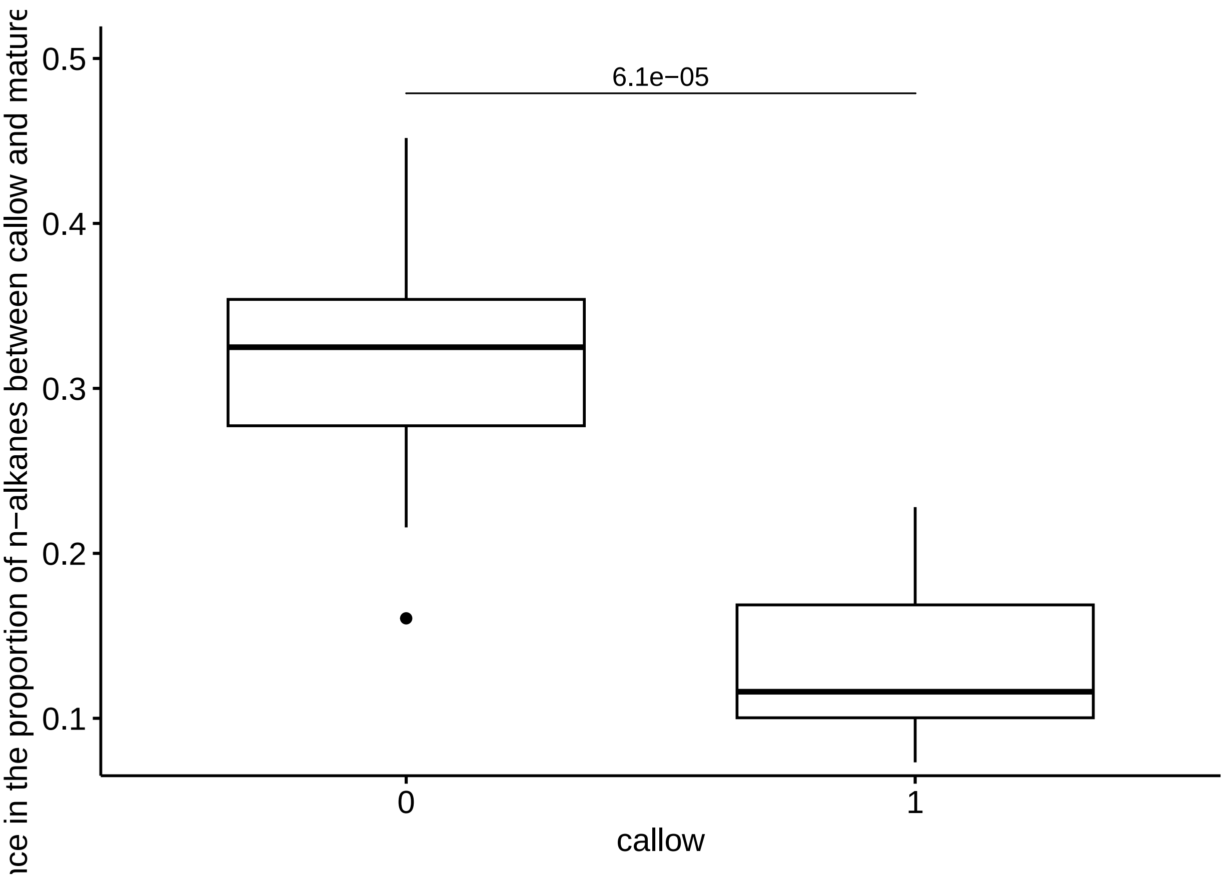
5.9 Difference in the proportion of CHC characteristic of *F. fusca* between callow *F. sanguinea* and *F. fusca*



5.10 Difference in the mass of *n*-alkanes between callow and mature *F. sanguinea*



5.11 Difference in the proprtion of *n*-alkanes between callow and mature *F. sanguinea*



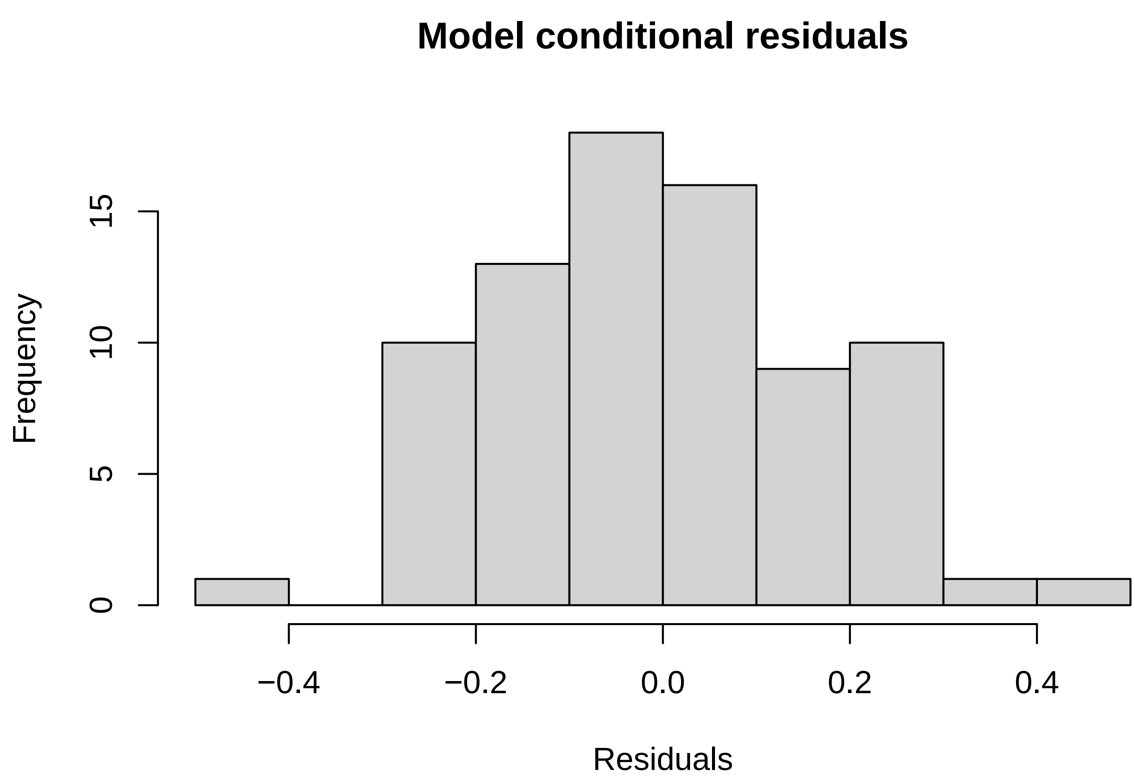
6 Change in the CHC profile in separated callow *F. sanguinea* ants

6.1 Change in the total CHC amount over time

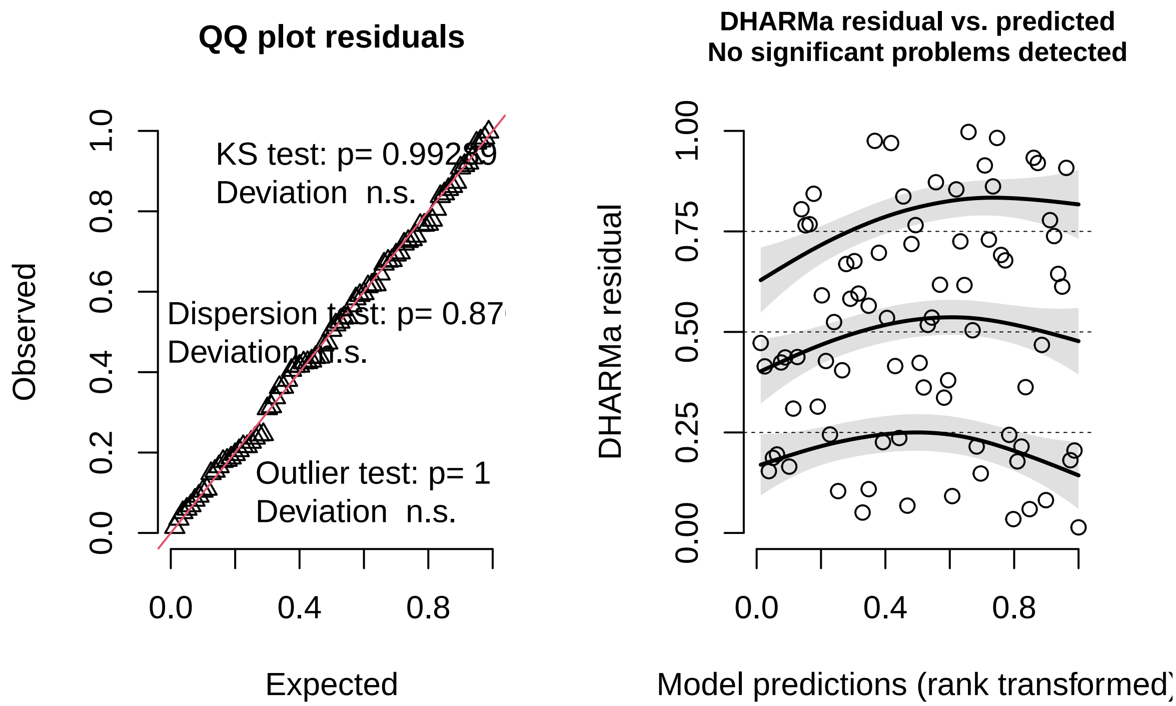
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(sqrt(corrected_mass)) ~ mean_delta + (1 | colony)
##    Data: separation_data[-74, ]
##
## REML criterion at convergence: -12.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.54596 -0.65494 -0.04106  0.63627  2.54062
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  colony   (Intercept) 0.01293  0.1137
##  Residual             0.03516  0.1875
## Number of obs: 79, groups:  colony, 11
```

```
##
## Fixed effects:
##              Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)  1.180332   0.047605 16.450772  24.794 1.83e-14 ***
## mean_delta   0.002458   0.001669 68.538937   1.473    0.145
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##            (Intr)
## mean_delta -0.518
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.99204, p-value = 0.9107
```



DHARMa residual

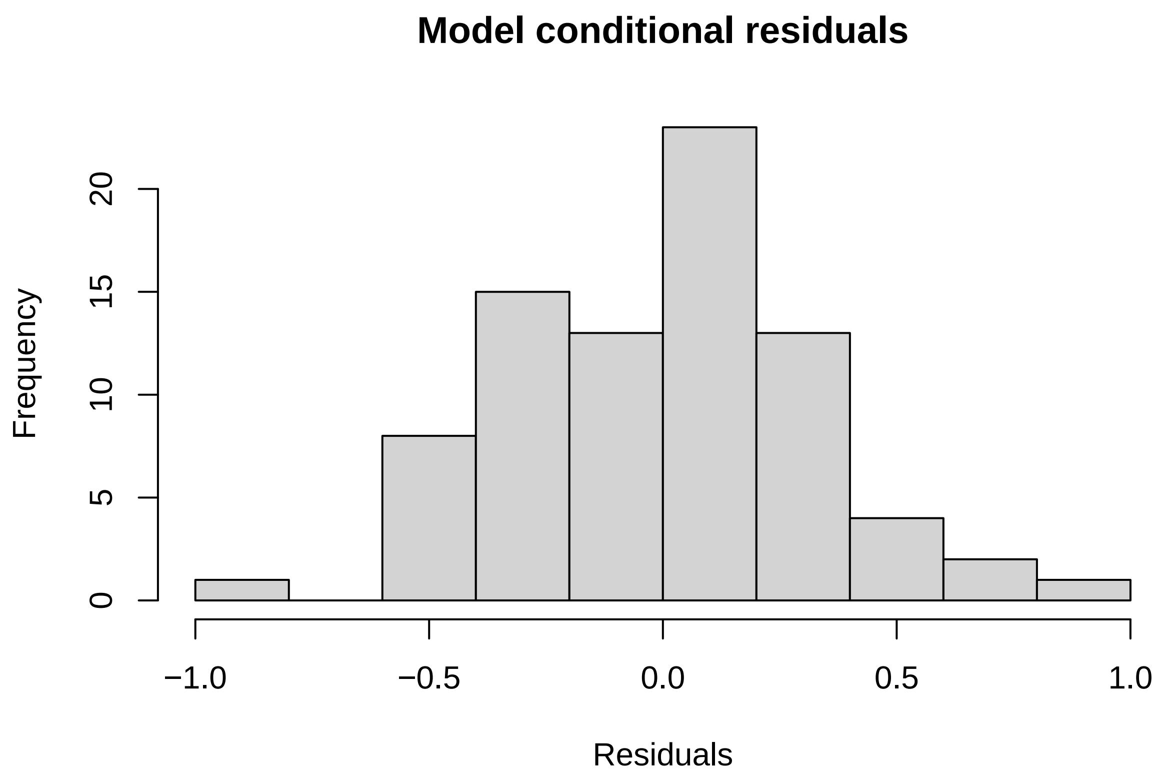


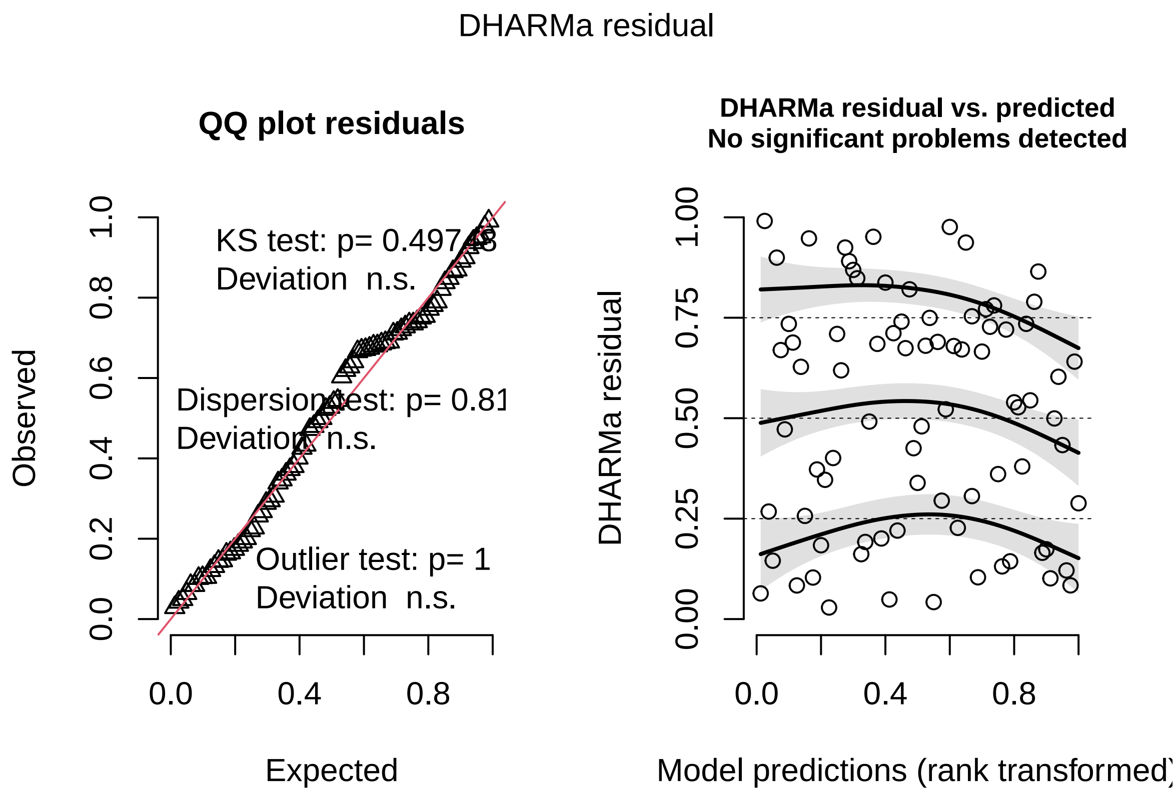
6.2 Change in the amount of compounds associated with callow *F. sanguinea*

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: log(corrected_mass_part) ~ mean_delta + (1 | colony)
##    Data: separation_data
##
## REML criterion at convergence: 87.5
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.4578 -0.6710  0.1080  0.5751  2.4611
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  colony   (Intercept) 0.08916  0.2986
##  Residual             0.11912  0.3451
## Number of obs: 80, groups:  colony, 11
##
## Fixed effects:
##              Estimate Std. Error        df t value Pr(>|t|)
## (Intercept) -0.370378   0.108556 13.530411  -3.412   0.0044 **
## mean_delta  -0.002148   0.002994 68.357201  -0.717   0.4755
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Correlation of Fixed Effects:
##            (Intr)
## mean_delta -0.417
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.99178, p-value = 0.8953
```



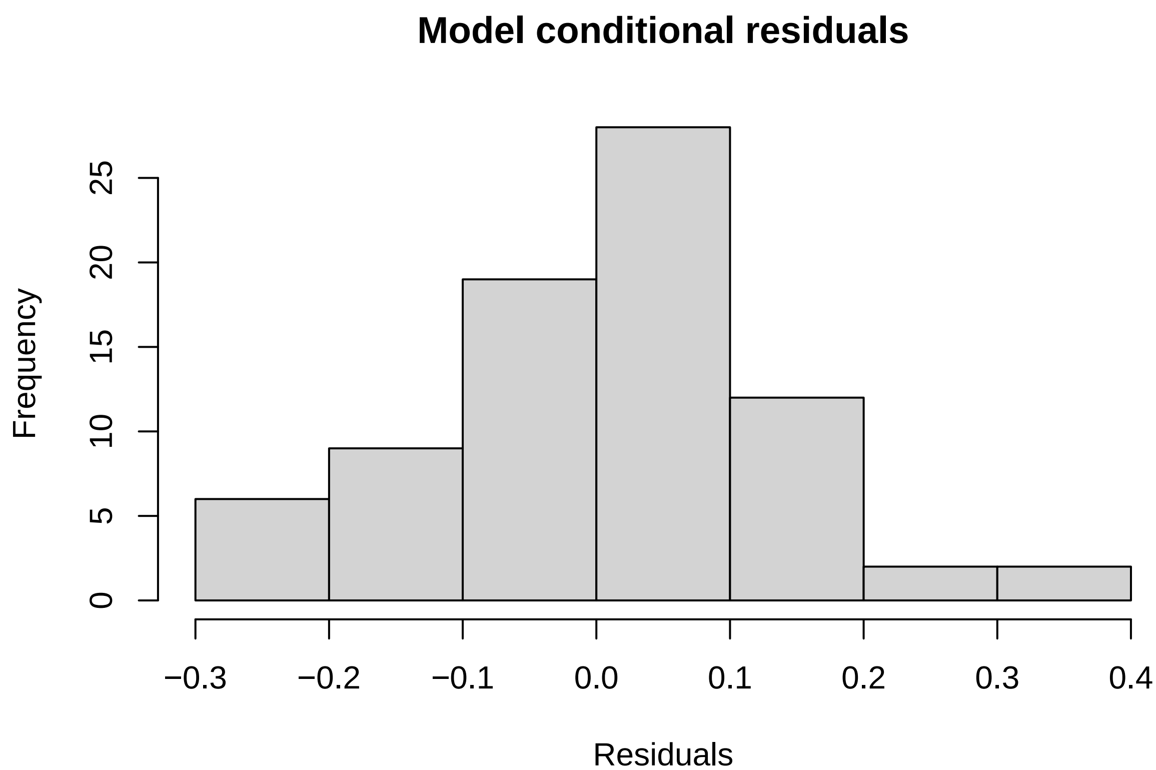


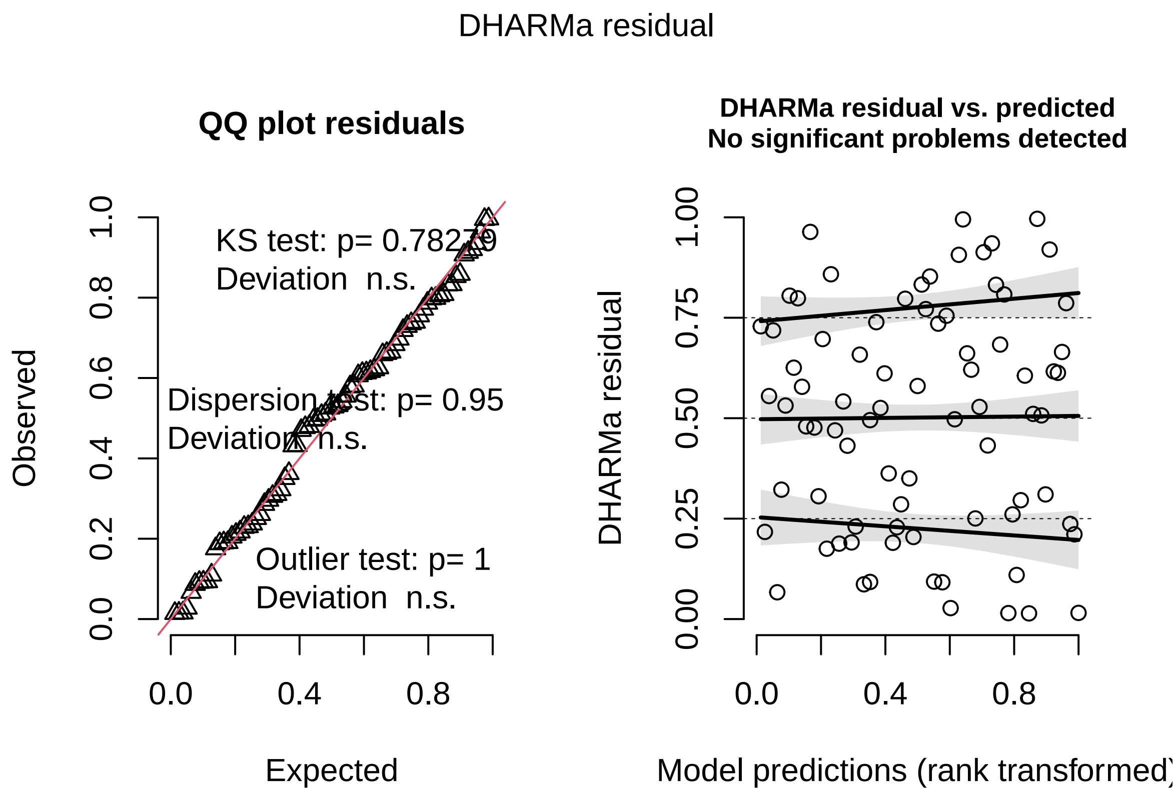
6.3 Change in the amount of compounds associated with *F. sanguinea*

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(sqrt(corrected_mass_part)) ~ poly(mean_delta, 1) + (1 | colony)
##    Data: separation_data[-c(69, 74), ]
##
## REML criterion at convergence: -86.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.22441 -0.61432  0.05969  0.71840  2.53565
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  colony   (Intercept) 0.00153  0.03912
##  Residual             0.01657  0.12874
## Number of obs: 78, groups:  colony, 11
##
## Fixed effects:
##                     Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)          0.53644    0.01904  7.80520  28.180 3.91e-09 ***
## poly(mean_delta, 1)  0.93763    0.13021 69.65336   7.201 5.49e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Correlation of Fixed Effects:
##              (Intr)
## ply(mn_d,1) 0.018

##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.98577, p-value = 0.5385
```



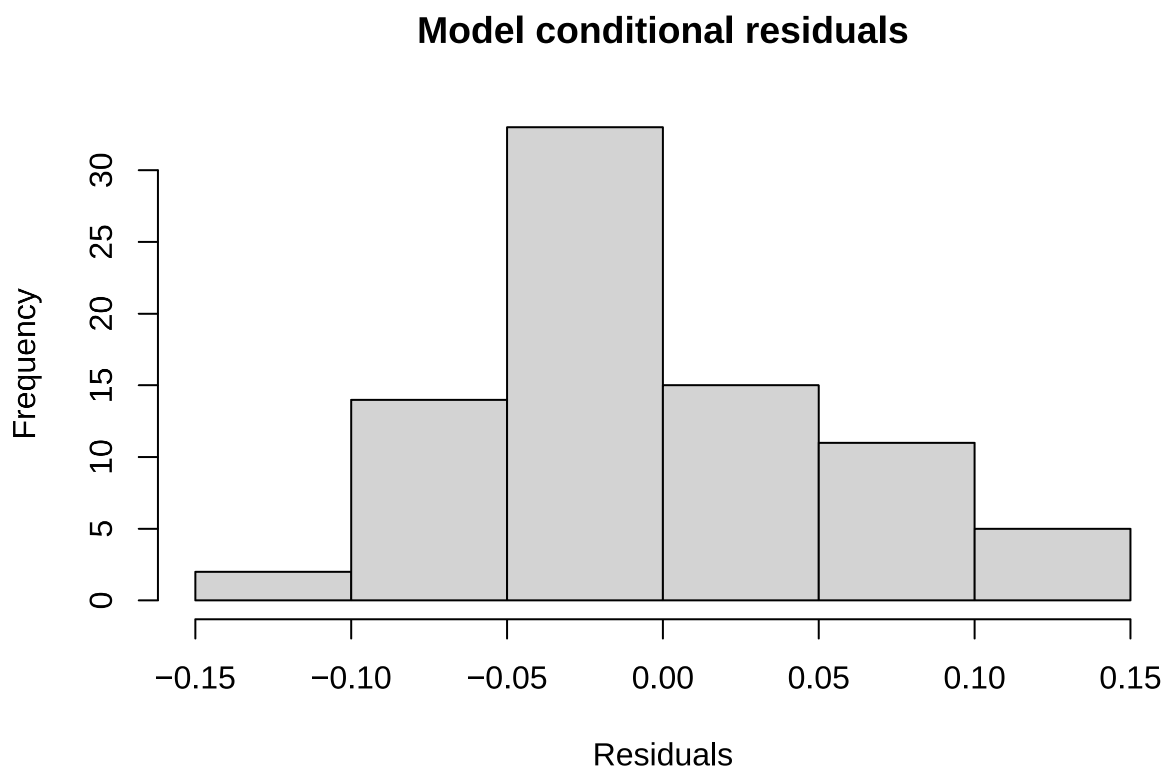


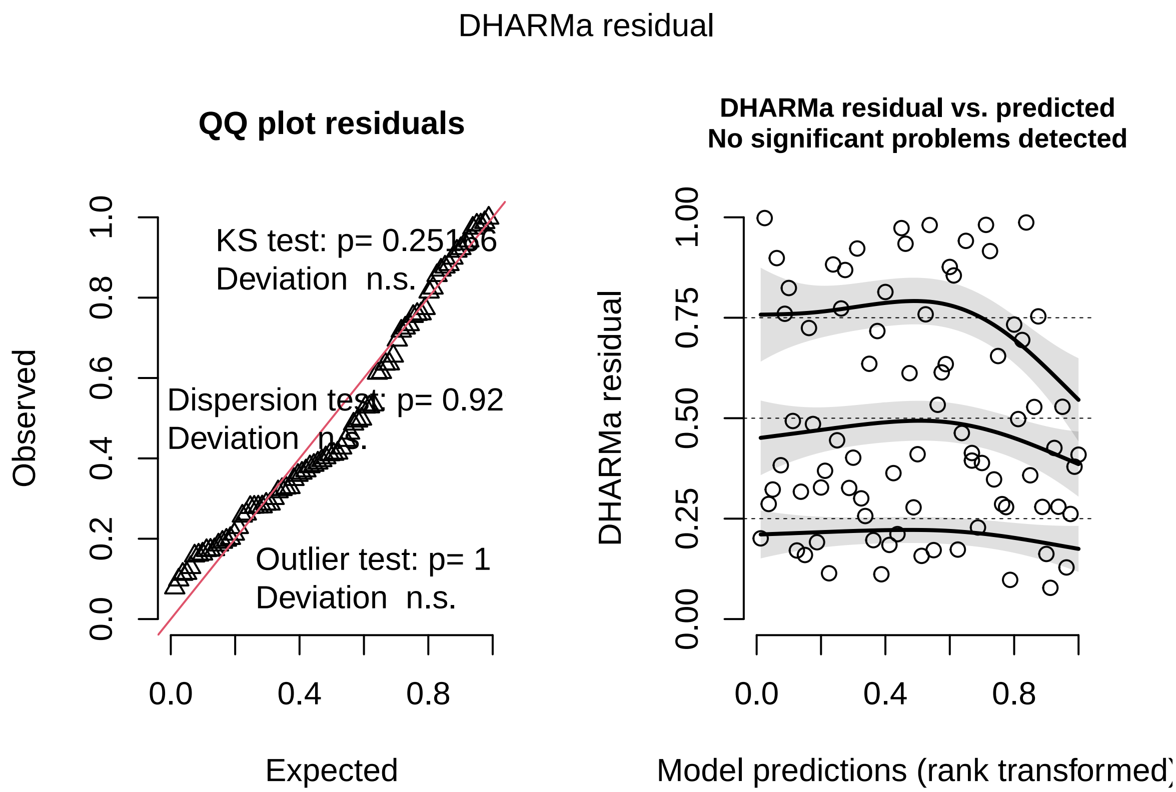
6.4 Change in the amount of compounds associated with *F. fusca*

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: corrected_mass_part ~ mean_delta + (1 | colony)
##    Data: separation_data
##
## REML criterion at convergence: -171.5
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.8982 -0.6538 -0.1157  0.6326  2.3286
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  colony   (Intercept) 0.006545 0.08090
##  Residual             0.003934 0.06272
## Number of obs: 80, groups:  colony, 11
##
## Fixed effects:
##               Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)  0.1739794  0.0267734 11.9815824   6.498 2.97e-05 ***
## mean_delta  -0.0017124  0.0005453 68.3928213  -3.140  0.00249 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Correlation of Fixed Effects:
##            (Intr)
## mean_delta -0.307

##
##  Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.96856, p-value = 0.04595
```





7 Impact of slave on the development of the *F. sanguinea* CHC profile

We analyzed the impact of *F. fusca* slaves on the CHC profile of callow *F. sanguinea* ants. In this experiment, *F. sanguinea* ants were isolated in pairs before eclosion, and their CHC profiles were analyzed at one of four time points marking their adult age. We then tested whether callow *F. sanguinea* ants were chemically more similar to their slave relatives from free-living colonies. Unrelated *F. fusca* ants served as a background group."

7.1 Analysis based on all samples

```
## Empirical p-value for the chemical distance difference for the ants aged 1-3 days:0.00000

## Number of colonies used in the treatment of 1-3 days of separation: 9

## Empirical p-value for the chemical distance difference for the ants aged 8-10 days:0.00000

## Number of colonies used in the treatment of 8-10 days of separation: 9

## Empirical p-value for the chemical distance difference for the ants aged 17-20 days:0.00080

## Number of colonies used in the treatment of 17-20 days of separation: 9
```

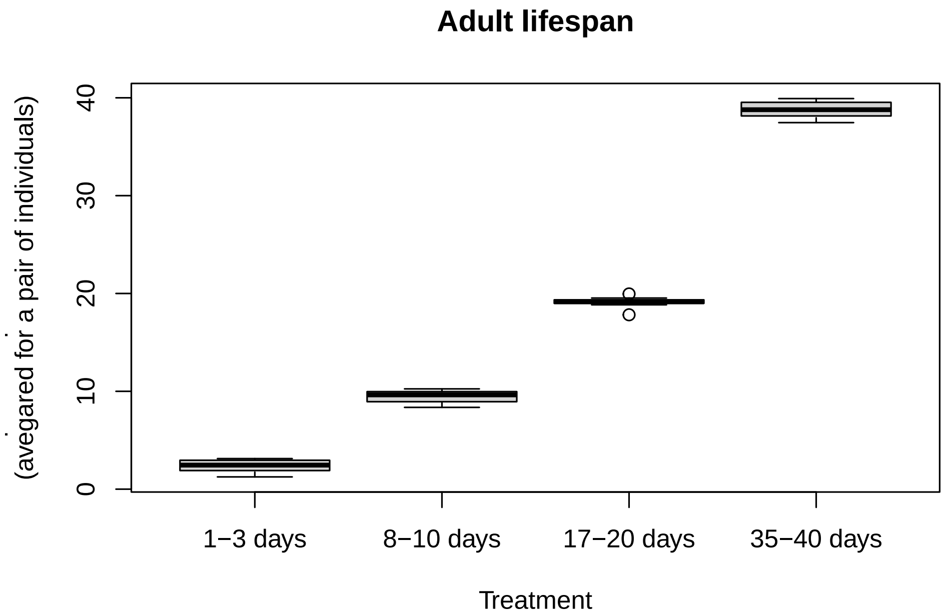


Figure S8 : Distribution of lifespans of ants used in the experiment. The precise age of individuals could not be determined because ants in pairs were not marked. Therefore, all possible combinations were considered to illustrate the potential range.

Table S3 : Chemical distance of the isolated *F. sanguinea* ant to *F. fusca* ants realted and unrelated to slaves.

Adult age	Chemical distance to slaves' relatives	Chemical distance to random colonies
1-3 days	0.3339	0.4723
8-10 days	0.3526	0.4809
17-20 days	0.4533	0.5296
35-40 days	0.5730	0.6101

```
## Empirical p-value for the chemical distance difference for the ants aged 35-40 days:0.00705
```

```
## Number of colonies used in the treatment of 35-40 days of separation: 8
```

7.2 Analysis restricted to the individuals hatched from cocoons

Since naked pupae might have acquired CHC through physical contact with adult ants after pupation, we subset our data to include only those samples which derived from the ants hatched from cocoons. In this case silky envelop during pupa stage ruled out possibility of the CHC transfer from the environment.

```
## Empirical p-value for the chemical distance difference for the ants aged 1-3 days:0.00000
```

```
## Number of colonies used in the treatment of 1-3 days of separation: 8
```

```
## Empirical p-value for the chemical distance difference for the ants aged 8-10 days:0.00001
```

Table S4 : Chemical distance of the isolated *F. sanguinea* ant to *F. fusca* ants realted and unrelated to slaves. Isolated ant spun a silky envelope before pupal stage, which prevented CHC transfer from the enviroment.

Adult age	Chemical distance to slaves' relatives	Chemical distance to random colonies
1-3 days	0.3480	0.4875
8-10 days	0.3752	0.4941
17-20 days	0.4655	0.5413
35-40 days	0.6176	0.6458

```
## Number of colonies used in the treatment of 8-10 days of separation: 8

## Empirical p-value for the chemical distance difference for the ants aged 17-20 days:0.00453

## Number of colonies used in the treatment of 17-20 days of separation: 8

## Empirical p-value for the chemical distance difference for the ants aged 35-40 days:0.03298

## Number of colonies used in the treatment of 35-40 days of separation: 7
```

8 Effect of dummy ants

In these experiments, *F. sanguinea* ants released from their cocoon envelopes were maintained in Petri dishes along with glass beads coated with CHC extracted from either *F. fusca* or *F. sanguinea* ants. We examined whether the contact with dummy nestmates influenced the development of CHC profile of young *F. sanguinea* workers. In particular, we aimed to determine there there is evidence for active chemical mimicry in *F. sanguinea*.

8.1 Distance to the CHC profile of the dummy ants treated by *F. fusca* CHCs

We computed the chemical distance to the CHC profile of ants whose CHCs were used to cover glass beads serving as dummy ants. In the control variant, we used the ants subjected to clean glass beads.

```
##
##  Wilcoxon signed rank exact test
##
## data:  to_test$mean_dist.x and to_test$mean_dist.y
## V = 19, p-value = 0.7344
## alternative hypothesis: true location shift is not equal to 0
```

8.2 Distance to the CHC profile of the dummy ants treated by *F. sanguinea* CHCs

```
##
##  Wilcoxon signed rank exact test
##
## data:  to_test$mean_dist.x and to_test$mean_dist.y
## V = 15, p-value = 0.2324
## alternative hypothesis: true location shift is not equal to 0
```

Table S5 : Chemical distance of separated *F. sanguinea* ants to the CHC profile of *F. fusca* ants from colonies that served as a source of CHC to caot the glass beads. In control variant, glass bead were left clean.

Colony ID	Treamtent	Averaged distance	Contrast	Averaged distance
SD18-3	12-15 days + F. fusca hydrocarbons	0.3957886	12-15 days (control)	0.5093071
SD18-5	12-15 days + F. fusca hydrocarbons	0.6157811	12-15 days (control)	0.5132649
SD18-5	12-15 days + F. fusca hydrocarbons	0.8423114	12-15 days (control)	0.8316840
SD19-11	12-15 days + F. fusca hydrocarbons	0.4665112	12-15 days (control)	0.4691791
SD19-3	12-15 days + F. fusca hydrocarbons	0.4996918	12-15 days (control)	0.4361776
SD19-4	12-15 days + F. fusca hydrocarbons	0.3049006	12-15 days (control)	0.4244113
SD19-6	12-15 days + F. fusca hydrocarbons	0.6735461	12-15 days (control)	0.6991620
SD19-8	12-15 days + F. fusca hydrocarbons	0.5954873	12-15 days (control)	0.5870741
SD20-2	12-15 days + F. fusca hydrocarbons	0.6985494	12-15 days (control)	0.7087854

Table S6 : Chemical distance of separated *F. sanguinea* ants to the CHC profile of *F. sanguinea* ants from colonies that served as a source of CHC to caot the glass beads. In control variant, glass bead were left clean.

Colony ID	Treamtent	Averaged distance	Contrast	Averaged distance
SD18-2	12-15 days + F. sanguinea hydrocarbons	0.5706616	12-15 days (control)	0.3734387
SD18-3	12-15 days + F. sanguinea hydrocarbons	0.4565024	12-15 days (control)	0.5986922
SD18-5	12-15 days + F. sanguinea hydrocarbons	0.5246184	12-15 days (control)	0.5800488
SD18-5	12-15 days + F. sanguinea hydrocarbons	0.2501310	12-15 days (control)	0.2323064
SD19-11	12-15 days + F. sanguinea hydrocarbons	0.5953242	12-15 days (control)	0.6070599
SD19-3	12-15 days + F. sanguinea hydrocarbons	0.4185158	12-15 days (control)	0.5112225
SD19-4	12-15 days + F. sanguinea hydrocarbons	0.6477936	12-15 days (control)	0.6680334
SD19-6	12-15 days + F. sanguinea hydrocarbons	0.5525058	12-15 days (control)	0.5908247
SD19-8	12-15 days + F. sanguinea hydrocarbons	0.8501448	12-15 days (control)	0.8499395
SD20-2	12-15 days + F. sanguinea hydrocarbons	0.6460560	12-15 days (control)	0.6543900

Table S7 : Proportion of *n*-docosane in CHC extracted from *F. sanguinea* ants maintained with the glass beads coated with the CHC of *F. fusca* ants and contaminated with *n*-docosane. In control variant, glass bead were left clean.

Colony ID	Treamtent	Averaged C22 fraction	Contrast	Averaged C22 fraction
SD18-3	12-15 days + F. fusca hydrocarbons	0.0178168	12-15 days (control)	0.0000000
SD18-5	12-15 days + F. fusca hydrocarbons	0.0012056	12-15 days (control)	0.0000000
SD18-5	12-15 days + F. fusca hydrocarbons	0.0017152	12-15 days (control)	0.0003274
SD19-11	12-15 days + F. fusca hydrocarbons	0.0112850	12-15 days (control)	0.0019864
SD19-3	12-15 days + F. fusca hydrocarbons	0.0086141	12-15 days (control)	0.0009720
SD19-4	12-15 days + F. fusca hydrocarbons	0.0173323	12-15 days (control)	0.0000000
SD19-6	12-15 days + F. fusca hydrocarbons	0.0091910	12-15 days (control)	0.0011031
SD19-8	12-15 days + F. fusca hydrocarbons	0.0026387	12-15 days (control)	0.0021076
SD20-2	12-15 days + F. fusca hydrocarbons	0.0086688	12-15 days (control)	0.0049489
W17-1	12-15 days + F. fusca hydrocarbons	0.0211912	12-15 days (control)	0.0014942

8.3 Fraction of *n*-docosane in the CHC profile

For the control treatment, glass beads were left uncoated with CHC. In the experimental treatments, the CHC used to coat the glass beads were supplemented with *n*-docosane, which served as an internal standard. We examined differences in the proportion of *n*-docosane in the CHC profiles of the tested ants, as these could indicate the acquisition of chemicals from the dummy ants.

8.3.1 *F. fusca*

```
##
##  Wilcoxon signed rank exact test
##
## data:  to_test$C22_prop.x and to_test$C22_prop.y
## V = 55, p-value = 0.001953
## alternative hypothesis: true location shift is not equal to 0
```

8.3.2 *F. sanguinea*

```
##
##  Wilcoxon signed rank exact test
##
## data:  to_test$C22_prop.x and to_test$C22_prop.y
## V = 63, p-value = 0.004883
## alternative hypothesis: true location shift is not equal to 0
```

Table S8 : Proportion of *n*-docosane in CHC extracted from *F. sanguinea* ants maintained with the glass beads coated with the CHC of *F. sanguinea* ants and contaminated with *n*-docosane. In control variant, glass bead were left clean.

Colony ID	Treamtent	Averaged C22 fraction	Contrast	Averaged C22 fraction
SD18-2	12-15 days + F. sanguinea hydrocarbons	0.0050660	12-15 days (control)	0.0016166
SD18-3	12-15 days + F. sanguinea hydrocarbons	0.0141292	12-15 days (control)	0.0000000

SD18-5	12-15 days + F. sanguinea hydrocarbons	0.0032893	12-15 days (control)	0.0000000
SD18-5	12-15 days + F. sanguinea hydrocarbons	0.0021351	12-15 days (control)	0.0003274
SD19-11	12-15 days + F. sanguinea hydrocarbons	0.0040419	12-15 days (control)	0.0019864
SD19-3	12-15 days + F. sanguinea hydrocarbons	0.0032399	12-15 days (control)	0.0009720
SD19-4	12-15 days + F. sanguinea hydrocarbons	0.0017680	12-15 days (control)	0.0000000
SD19-6	12-15 days + F. sanguinea hydrocarbons	0.0035583	12-15 days (control)	0.0011031
SD19-8	12-15 days + F. sanguinea hydrocarbons	0.0017423	12-15 days (control)	0.0021076
SD20-2	12-15 days + F. sanguinea hydrocarbons	0.0043650	12-15 days (control)	0.0049489
W17-1	12-15 days + F. sanguinea hydrocarbons	0.0030204	12-15 days (control)	0.0014942

9 Body surface area of *F. sanguinea* ants and their slaves

In the publication, CHC amounts are reported relative to the square of head width. To assess whether this measure reliably reflects body size, we measured the planar projection areas of different body parts of *F. sanguinea* and *F. fusca*. The plots below suggest that head width slightly underestimates body surface area in *F. fusca* compared to *F. sanguinea*. As a result, the relative concentration of CHC in *F. fusca* may be overestimated. However, this bias acts in the opposite direction of our findings—*F. sanguinea* exhibits greater CHC concentration on its cuticle. Therefore, the shape-related bias between species is conservative, as it reduces the likelihood of rejecting the null hypothesis

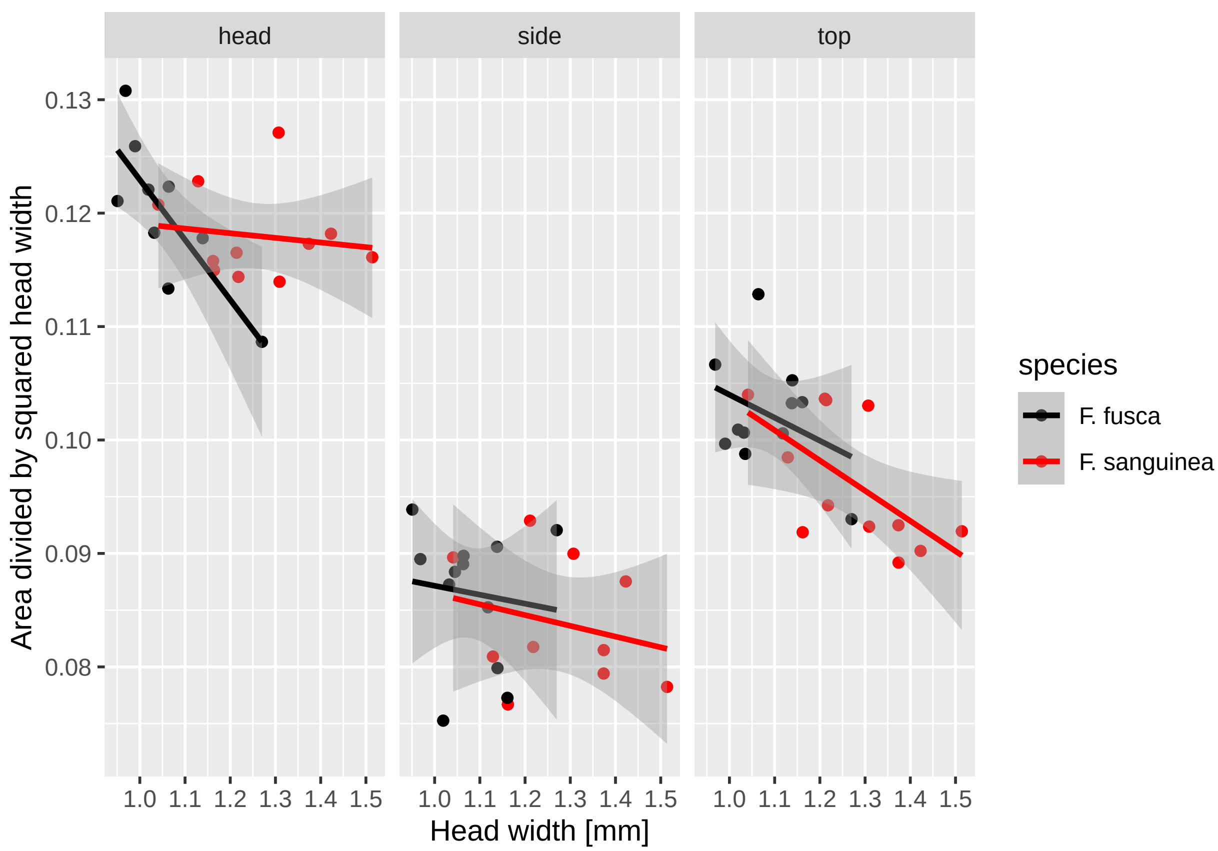


Figure S9 : Ratios of the planar projection areas of three body parts (head, thorax dorsal view, thorax lateral view) to the square of head width.

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